1) Calculate evaluation metrics for the following (31) A Confusion matrix of a binary classifier

	Predicted Positive	Predicted Negative
Actual Positive	50 tp	10 fn
Actual Negative	5 fp	100 tn

$tp = 50 \quad tn = 100 \quad fp = 5 \quad fn = 10$

$$\text{Accuracy} = \frac{150}{50+100+10+5} = \frac{150}{165} = 0.9090$$

$$\text{Precision} = \frac{tp}{tp+fp} = \frac{50}{50+5} = \frac{50}{55} = 0.9090$$

$$\text{Recall} = \frac{tp}{tp+fn} = \frac{50}{50+10} = \frac{50}{60} = 0.8333$$

$$\text{Specificity} = \frac{tn}{tn+fp} = \frac{100}{100+5} = 0.9523$$

2) Consider a 3-class classification problem with following confusion matrix, Calculate accuracy, precision and recall of class A, class B and class C. Also F1 Score of Class A, B, C

[contd. in next page]

(31) B

Actual \ Predicted	A	B	C
A	40	5	5
B	10	30	10
C	5	10	35

$$\text{Accuracy} = \frac{40+30+35}{40+30+35+5+5+10+10+5+10} = \frac{105}{150} = 0.7$$

$$P(A) = \text{Precision}(A) = \frac{tp}{tp+fp} = \frac{40}{40+15} = 0.7272$$

	A	B	C
A	40	5	5
B	10	30	10
C	5	10	35

$$R(A) = \text{Recall}(A) = \frac{tp}{tp+fn} = \frac{40}{40+10} = \frac{40}{50} = 0.8$$

$$\text{F1 Score} = 2 \times \frac{P(A)R(A)}{P(A)+R(A)} = \frac{2 \times 0.7272 \times 0.8}{0.7272+0.8} = 0.7618$$

$$\text{Precision}(B) = \frac{tp}{tp+fp}$$

	A	B	C
A	40	5	5
B	10	30	10
C	5	10	35

$$\text{Precision}(B) = \frac{30}{30+15} = 0.6667$$

$$\text{Recall}(B) = \frac{tp}{tp+fN} = \frac{30}{30+20} = 0.6$$

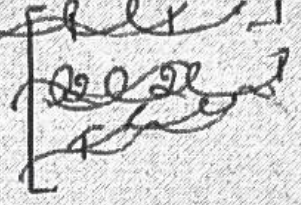
$$\text{F1 Score}(B) = \frac{2 \times 0.6667 \times 0.6}{0.6667+0.6} = 0.6315$$

$$\text{Precision}(C) = \frac{tp}{tp+fp} = \frac{35}{35+15} = 0.7$$

$$\text{Recall}(C) = \frac{tp}{tp+fN} = \frac{35}{35+15} = 0.7$$

$$\text{F1 Score}(C) = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times 0.7 \times 0.7}{0.7+0.7} = 0.7$$

accuracy and
Calculate precision, recall and f1-score for each class
for the following confusion matrix



Calculate

1) For the following confusion matrix, calculate (i) & (ii)
   (i) Accuracy.
   (ii) Precision, Recall, F1-Score for class-A

Solution:

Actual \ Predicted	A	B	C
A	1	1	1
B	2	2	1
C	1	2	4

(i) Accuracy $= \frac{1+2+4}{15} = 0.4667$

(ii) class A: precision (A) $= \frac{tp}{tp+fp} = \frac{1}{1+3} = 0.25$

	tp	fn
	1	1
	2	2	1
	1	2	4
	fp	tn

$\text{Recall (A)} = \frac{tp}{tp+fn} = \frac{1}{1+2} = 0.3333$

$\text{F1 Score} = \frac{2 \times \text{precision (A)} \times \text{Recall (A)}}{\text{precision (A)} + \text{Recall (A)}} = 0.2857$

2) Repeat the above problem for class B or (ii) for class B

$\text{precision (B)} = \frac{tp}{tp+fp} = \frac{2}{2+3} = 0.4$

$\text{Recall (B)} = \frac{tp}{tp+fn} = \frac{2}{2+3} = \frac{2}{5} = 0.4$

$\text{F1-Score} = \frac{2 \times \text{Precision (B)} \times \text{Recall (B)}}{\text{precision (B)} + \text{Recall (B)}} = \frac{2 \times 0.4 \times 0.4}{0.4 \times 0.4} = 0.4$

(iii) Repeat Q(i) of the above problem

$$\text{precision (c)} = \frac{tp}{tp+fp}$$

$$A \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ 1 & 2 & 4 \end{bmatrix}$$

$$= \frac{4}{4+2} = 0.6667$$

$$\text{Recall (c)} = \frac{tp}{tp+fn} = \frac{4}{4+3} = 0.5714$$

$$\text{F-1 Score} = \frac{2 \times \text{precision (c)} \times \text{Recall (c)}}{\text{precision (c)} + \text{Recall (c)}}$$

$$= 0.6153$$

1) A Company wants to estimate the average time customers spend on its website daily. A random sample of 50 customers daily time on the website (in minutes) is collected. The sample mean is 35 minutes and the standard deviation of the sample is 8 minutes. The company wants 95% of confidence interval for the true population mean.

Solution: We know $P\{-1.96 < Z < 1.96\} = 0.95$ [From Z-table]

$$P\left\{-1.96 < \frac{(m-\mu)}{\frac{\sigma}{\sqrt{N}}} < 1.96\right\} = 0.95$$

$$P\left\{\left(m - 1.96 \times \frac{\sigma}{\sqrt{N}}\right) < \mu < \left(m + 1.96 \times \frac{\sigma}{\sqrt{N}}\right)\right\} = 0.95$$

Sample mean $m = 35$

std dev $\sigma = 8$

$N = 50$

$$m - 1.96 \times \frac{\sigma}{\sqrt{N}} = 35 - 1.96 \times \frac{8}{\sqrt{50}} = 32.7825$$

$$m + 1.96 \times \frac{\sigma}{\sqrt{N}} = 35 + 1.96 \times \frac{8}{\sqrt{50}} = 37.2174$$

Confidence Interval $= \left(m - 1.96 \times \frac{\sigma}{\sqrt{N}},\ m + 1.96 \times \frac{\sigma}{\sqrt{N}}\right)$

$= (32.7825, 37.2174)$

Inference: With 95% Confidence, the true average time customers spend on the Webside daily is between 32.7825 minutes and 37.2174 minutes.

This is from Research Methodology C.R. Kothari Page No 207

Example 10.12: A sample of 400 male students is found to have a mean height 67.47 inches. Can it be reasonably regarded as a sample mean from a large population with mean height 67.39 inches and standard deviation 1.30 inches? Test at 5% level of significance.

Solution:

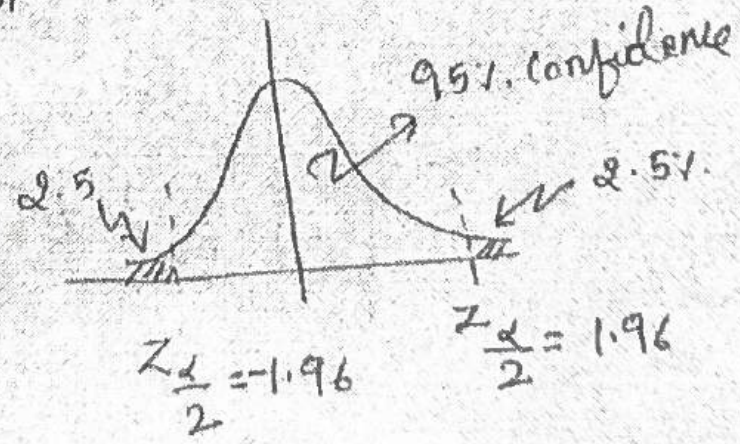
Taking the null hypothesis that the mean height of the population is equal to 67.39 inches, we can write

$H_0: \mu = 67.39$ [Null Hypothesis]

$H_1: \mu \neq 67.39$ [Alternative hypotheses]

$$Z = \frac{m - \mu}{\frac{\sigma}{\sqrt{N}}} = \frac{67.47 - 67.39}{\frac{1.3}{\sqrt{400}}}$$

$$Z = 1.231$$



Since the observed value of $Z = 1.231$ is in the acceptance region, H_0 is accepted

Example: 10.13) The mean of a certain production process is known to be 50 with a standard deviation of 2.5. The production manager may welcome any change in mean value towards higher side but would like to safeguard against decreasing values of mean. He takes a sample of 120 items that gives a mean value of 48.5. What interference should the manager take for the production process on the basis of sample results? use 5 per cent level of significance for the purpose.

Solution: Taking the mean value of the population to be 50, we may write

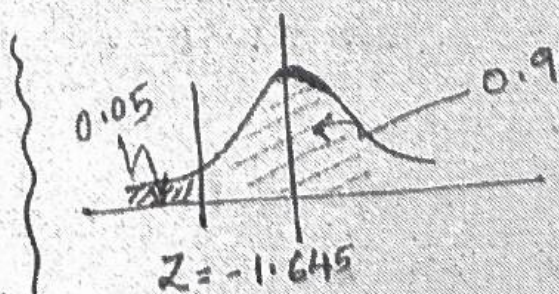
$H_0: \mu = 50$

$H_1: \mu < 50$ (since the manager wants to safeguard against decreasing values of mean)

$$Z = \frac{m - \mu_0}{\frac{\sigma}{\sqrt{N}}} = \frac{48.5 - 50}{\frac{2.5}{\sqrt{120}}}$$

$Z = -6.57426$

As H_1 is one sided in the given question, we shall determine the rejection region applying one-tailed test at 5% level of significance.



The observed value of $Z = -6.57426$ is in the rejection region and thus the level of Ho is rejected at 5% significance level. We can conclude that the production process is showing mean which is significantly less than the population mean and this calls for some corrective action concerning the said process

R. Mby Kothari

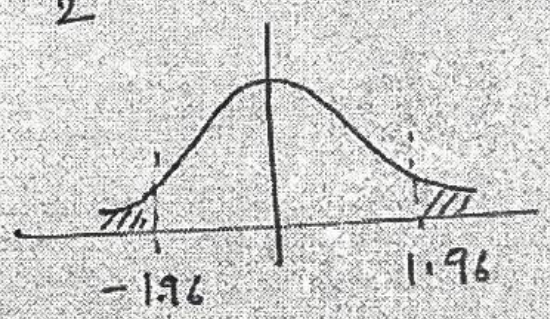
Example 9.9) From a random sample of 36 New Delhi civil service personnel, the mean age and standard deviation were found to be 40 years and 4.5 years respectively. Construct a 95% confidence interval for the mean age of civil servants in New Delhi.

Solution:

$$N = 36$$
$$m = 40$$
$$\sigma = 4.5$$

$$P\left\{ m - z_{\frac{\alpha}{2}} \left(\frac{\sigma}{\sqrt{N}}\right) < \mu < m + z_{\frac{\alpha}{2}} \text{ and } \frac{\sigma}{\sqrt{N}} \right\}$$

95% Confidence $\Rightarrow z_{\frac{\alpha}{2}} = 1.96$



$$m - 1.96 \times \frac{\text{and } \sigma}{\sqrt{N}}$$

$$= m - 1.96 \times \frac{4.5}{\sqrt{36}}$$

$$= 40 - 1.47$$

$$= 38.53$$

$$m + 1.96 \times \frac{\sigma}{\sqrt{N}}$$

$$= m + 1.96 \times \frac{4.7}{\sqrt{36}}$$

$$= 40 + 1.47$$

$$= 41.47$$

$\therefore$ Confidence Interval = (38.53, 41.47)

1) A candy machine makes chacollate bar that are average 5 gm. A worker claims that the machine after maintenance no longer makes 5 gm bars. Write the Null Hypothesis and alternative hypothesis.

Solution

$H_0 : \mu = 5\,gm$ (Null Hypothesis)

$H_1 : \mu \neq 5\,gm$ (Alternate Hypothesis)

---

2) A machine dispenses 80 mL of fluid in a bottle on average. An employer studies 40 samples and finds the sample mean to be 78 mL. Assume $\sigma = 2.5$. Design a hypothesis test such that the significance level is 0.05.

Solution :-

$H_0 : \mu = 80$ Null Hypothesis

$H_1 : \mu \neq 80$ Alternate hypothesis

Significance level = 0.05

Confidence level = 1 − 0.05 = 0.95

$m = 78$

$\sigma = 2.5$

$\mu = 80$

$N = 40$

$$Z = \frac{m - \mu_0}{\frac{\sigma}{\sqrt{N}}} = \frac{78 - 80}{\frac{2.5}{\sqrt{40}}}$$

$Z = -5.059$

2) A factory has a machine that dispenses 80mL of fluid in a bottle. An employee believes the average amount of fluid is not 80mL. Using 40 Samples, he measures the average amount dispensed by the machine to be 78mL with a Standard deviation of 2.5

(a) State the null and Alternate Hypothesis

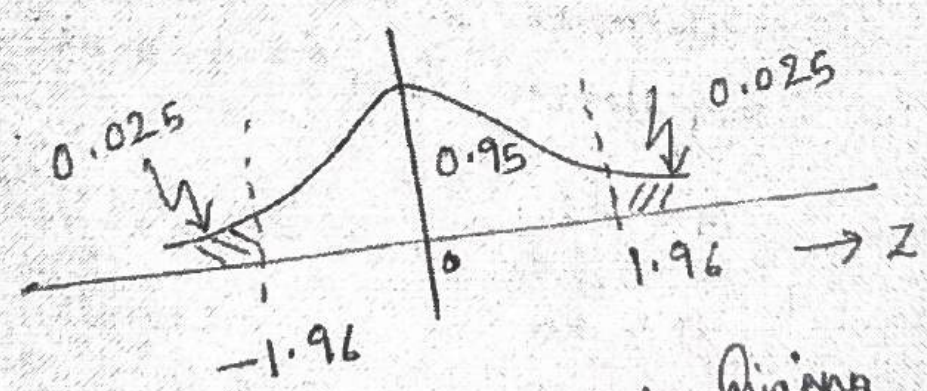
(b) At a 95% Confidence level, is there enough evidence to support the idea that machine is not working properly.

Solution: $H_0 : \mu = 80$ (Null Hypothesis)

$H_1 : \mu \neq 80$ (Alternate Hypothesis)

$$Z = \frac{m - \mu}{\frac{\sigma}{\sqrt{N}}} = \frac{78 - 80}{\frac{2.5}{\sqrt{40}}}$$

$$Z = -5.059$$



Since Z is not lie between -1.96 and 1.96, we can not accept Null Hypothesis. There is enough evidence to support the idea that machine is not working properly

Z does not lying

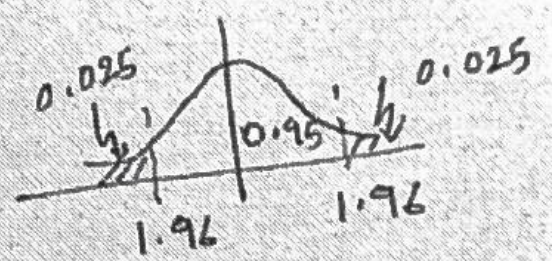
3) The average test scores in a physics class is normally distributed with a standard deviation of 5.4. 50 scores with a sample mean of 79 were selected at random.

(a)(a) Find a 95% confidence interval for the population mean test score

(b) what is the value of the margin error?

Solution: $m = 79$ $\sigma = 5.4$ $N = 50$

95% confidence interval



$$Z = \frac{m - \mu}{\frac{\sigma}{\sqrt{N}}}$$

$$P\left\{ m - 1.96\frac{\sigma}{\sqrt{N}} \leq \mu \leq m + 1.96\frac{\sigma}{\sqrt{N}} \right\}$$

$$m - 1.96 \times \frac{5.4}{\sqrt{50}} = 79 - 1.96\frac{5.4}{\sqrt{50}} = 77.5031$$

$$m + 1.96 \times \frac{5.4}{\sqrt{50}} = 79 + 1.96 \times \frac{5.4}{\sqrt{50}} = 80.4968$$

Confidence Interval = $(77.5031, 80.4968)$

Value of the margen error = $1.96 \times \frac{\sigma}{\sqrt{N}}$ (Z)

$= 1.96 \times \frac{5.4}{\sqrt{50}}$

$= \underline{1.4968}$

Example 9.11 : The foreman of ABC mining company has estimated the average quantity of iron ore extracted to be 36.8 tons per shift and the standard deviation to be 2.8 tons per shift, based upon a random selection of 4 shifts. Construct a 90 percent Confidence Interval around this estimate.

Solution : Size of the sample = 4 is small, we shall use t-distribution for finding required confidence interval about the population Mean.

$$m = 36.8$$

$$S_{n-1} = 2.8$$

$$N = 4$$

$$P\left\{ m - t_{\left(\frac{\alpha}{2}, N-1\right)} \frac{S}{\sqrt{N}} < \mu < m + t_{\frac{\alpha}{2}, N-1} \frac{S}{\sqrt{N}} \right\} = 1-\alpha$$

$$\mu > m - t_{\left(\frac{\alpha}{2}, N-1\right)} \frac{S}{\sqrt{N}}$$

$$\mu > 36.8 - (2.353) * \frac{2.8}{\sqrt{4}}$$

$$\mu < 36.8 + 2.353 * \frac{2.8}{\sqrt{4}}$$

$$\mu > 33.5058$$

$$\mu < 40.0942$$

Confidence Interval $= 36.8 \pm 3.2942$

$= (33.5058, 40.0942)$

Example 10.11) A certain chemical process is said to have produced 15 or less pounds of waste material for every 60 lbs. batch with a corresponding standard deviation of 5 lbs. A random sample of 100 batches gives an average of 16 lbs of waste per batch. Test at 10 percent level whether the average quality of waste per batch has increased. Compute the power of the test for $\mu = 16$ lbs. If we raise the level of significance to 20 percent, then how the power of the test for $\mu = 16$ lbs would be affected.?

Solution: As we want to test the hypothesis that average quantity of waste per batch of 60 lbs. is 15 or less pounds against the hypothesis that the waste quantity is more than 15 lbs, we can write

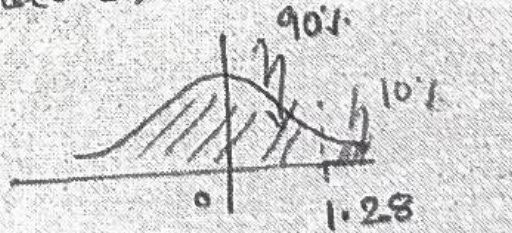
$H_0 : \mu \leq 15$ lbs (Null hypothesis)

$H_1 : \mu > 15$ lbs (Alternate hypothesis)

As H_1 is one-sided, we shall use the one-tailed test (in the right tail because H_1 is of more than type) at 10% level

$$m \leq \mu + 1.28 \frac{\sigma}{\sqrt{N}}$$

(z)



Give $\sigma = 5$

$N = 100$

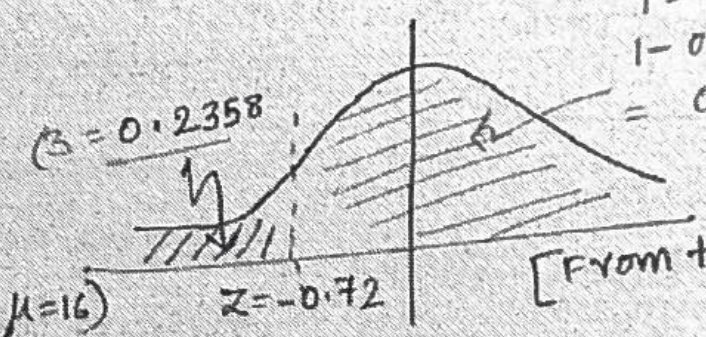
$$m \leq 15 + 1.28 \times \frac{5}{\sqrt{100}}$$

$$m \leq 15.64$$

$m \leq 15.64$ at 10% level of significance for accepting.

But the sample average is 16 lbs ($\mu = 16$ lbs), which does not come in the acceptance region as above, we there fore rejected Ho and conclude that average quantity of waste per batch has increased.

Now $$Z = \frac{m - \mu}{\frac{\sigma}{\sqrt{N}}} = \frac{15.64 - 16}{\frac{5}{\sqrt{100}}} = -0.72$$



1 − β = Power of the test
1 − 0.2358
= 0.7642

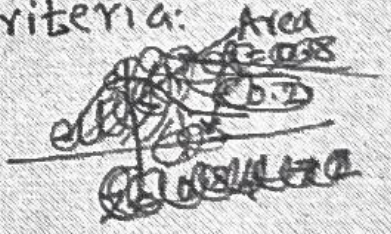
Since it is one tailed test

$\beta(\mu) = P_\mu\{m \leq 15.64 \mid \mu = 16\}$ [From two tailed Z-table]

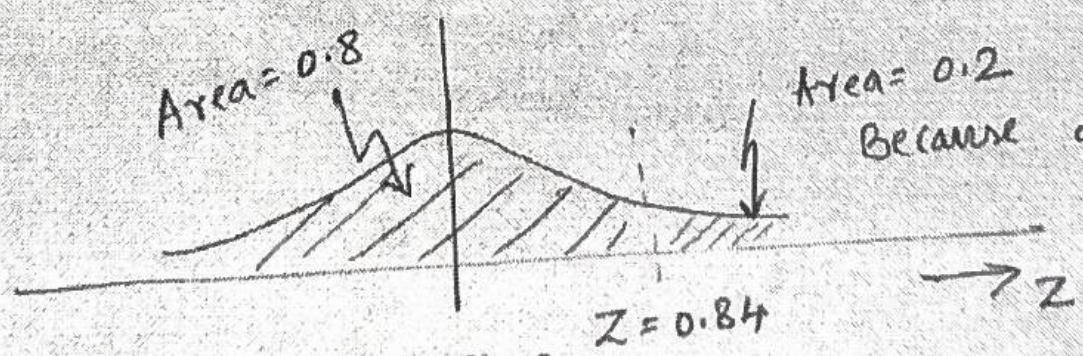
Power of the test = $1 - \beta = 1 - 0.2358 = 0.7642$

In other case of

In case the Significance level is raised to 20%, then we shall follow the following criteria:



Accept Ho if $m < \mu + Z \times \frac{5}{\sqrt{100}}$



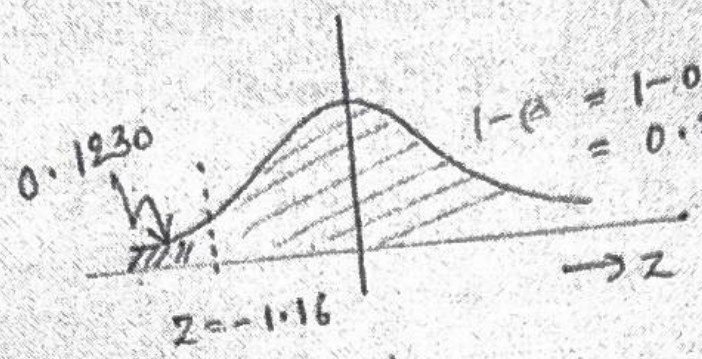
Because $\alpha = \frac{20}{100} = 0.2$

$\therefore m \leq 15 + 0.84 \times \frac{5}{\sqrt{100}}$

$m \leq 15.42$

m should be less than or equal to 15.42 for accepting

$$Z = \frac{m - \mu}{\frac{\sigma}{\sqrt{N}}} = \frac{15.42 - 16}{\frac{5}{\sqrt{100}}} = -1.16$$



∴ Power of the test = 1 − β

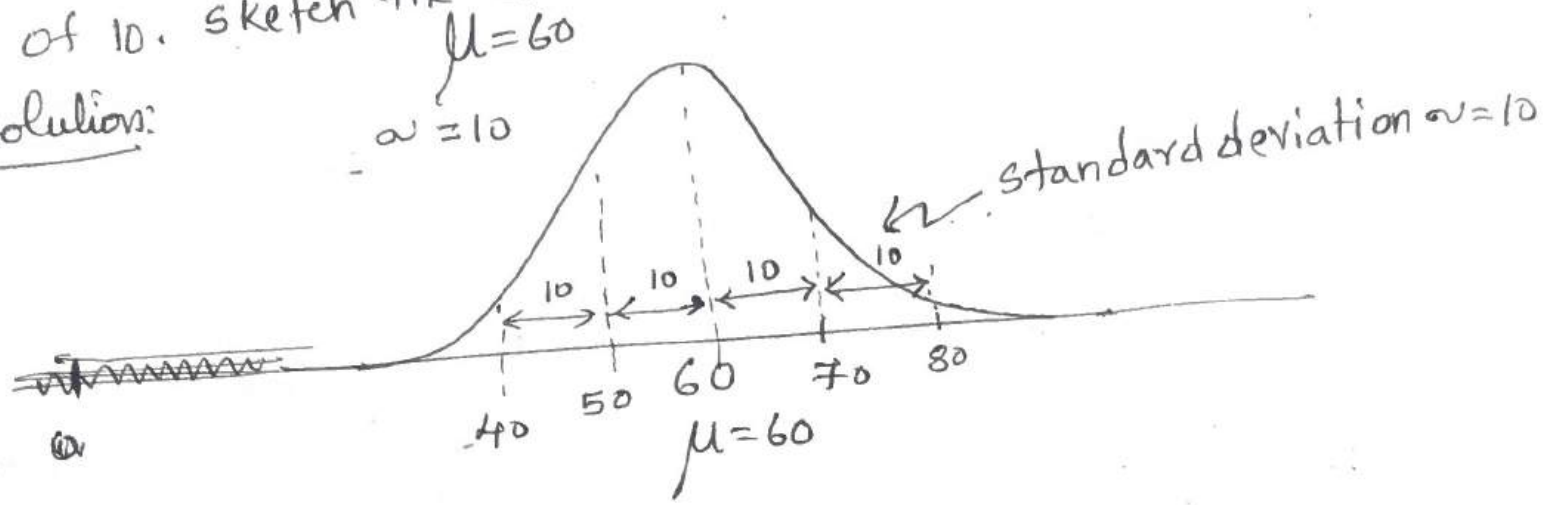
= 1 − 0.1230

= 0.8770

Examples of Standard Normal Distribution and z-table

(1) In an exam, the marks scored by the students follow normal distribution with mean of 60 and standard deviation of 10. sketch the distribution.

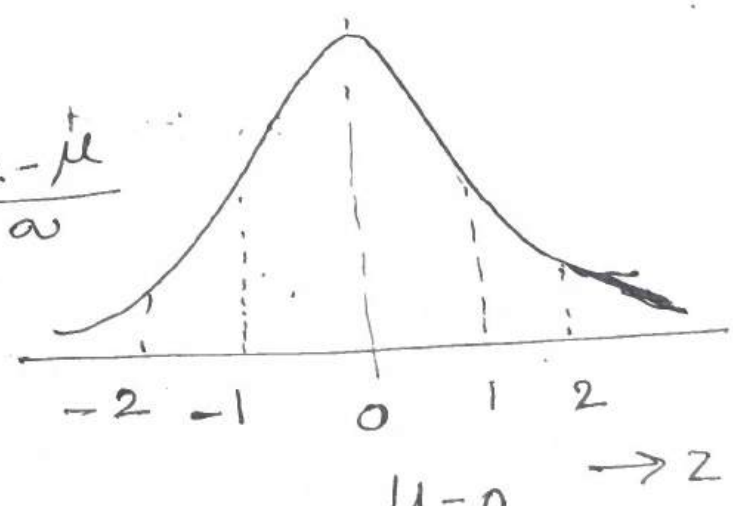
Solution:



(2) Transform the above distribution to standard normal distribution with $\mu = 0$ $\sigma = 1$

$\mu = 60$, $\sigma = 10$

$$Z = \frac{x - \mu}{\sigma} = \frac{x - 60}{\sigma} = \frac{x - 60}{10}$$



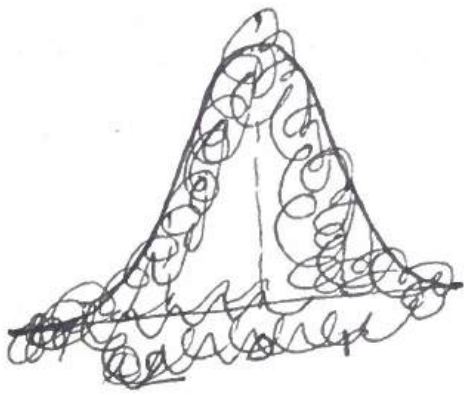
When $x = 60$ $Z = \frac{60-60}{10} = 0$

$x = 70$ $Z = \frac{70-60}{10} = 1$

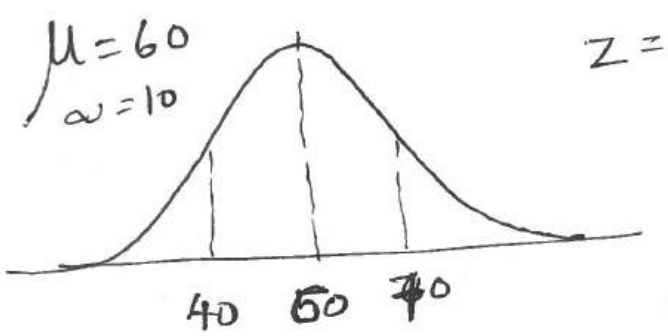
$x = 50$ $Z = \frac{50-60}{10} = -1$

$x = 40$ $Z = \frac{40-60}{10} = -2$

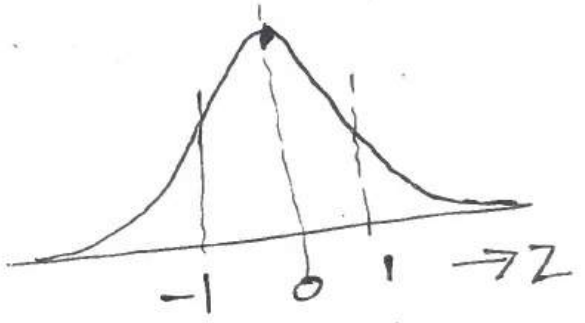




3) In an exam marks scored by students follows normal distribution with a mean of 60 and standard deviation of 10. What proportion of students scored less than 49 in this exam



$$Z = \frac{x-\mu}{\sigma}$$



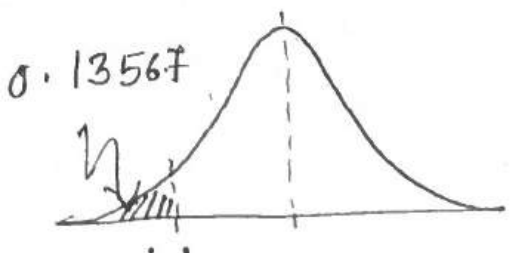
40 50 60

$\left(\frac{40-50}{10}\right)$ $\left(\frac{60-50}{10}\right)$

$$Z = \frac{x-\mu}{\sigma}$$

$$Z = \frac{49-60}{10}$$

$$= \frac{-11}{10} = -1.1$$



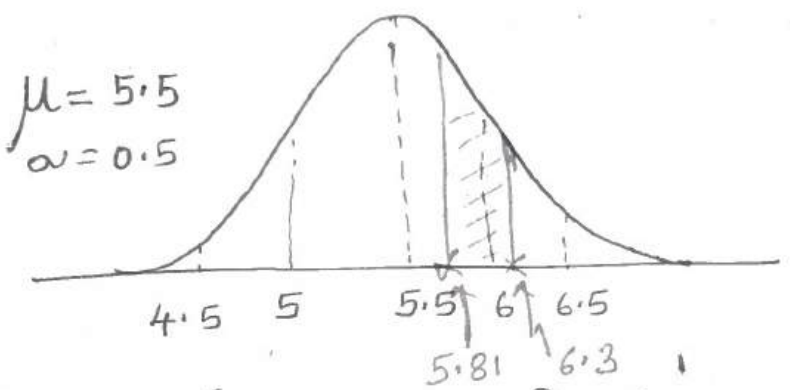
$P(Z < -1.1) = 0.13567$ [From Z-Table]

$P(Z < -1.1) \Rightarrow 13.567\%$ of students have got less than 49

$\therefore \quad P(X < 49) = 13.567\%$

24) When measuring the height of all the students of a School, it was found that it ~~was~~ is normally distributed with a mean height 5.5 feet and standard deviation of 0.5 feet. What proportion of students are between 5.81 feet and 6.3 feet tall?

Solution:



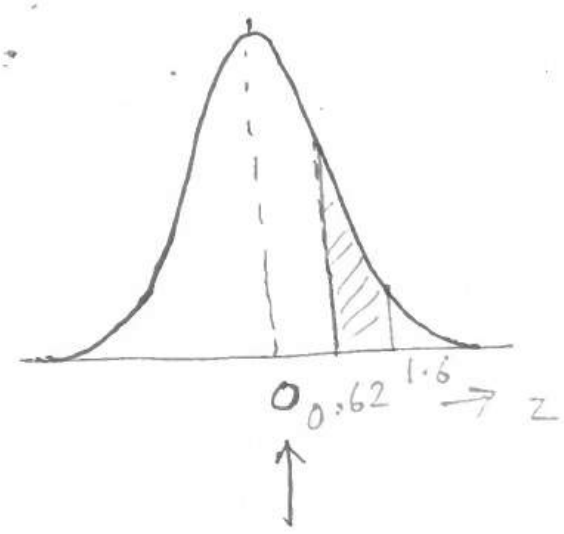
$P(5.8 < X < 6.3) = ?$

$$Z = \frac{X - \mu}{\sigma}$$

$$Z = \frac{X - 5.5}{0.5}$$

$X = 5.81$ $\quad Z = \frac{5.81 - 5.5}{0.5} = 0.62$

$X = 6.3$ $\quad Z = \frac{6.3 - 5.5}{0.5} = 1.6$



$P(Z < 0.62) = 0.73237$ [From z-Table]

$P(Z < 1.6) = 0.94520$ [From z-Table]

$P(0.62 < Z < 1.6) = 0.94520 - 0.73237$

$= 0.21283$

$\Rightarrow$ 21.283% of students are between 5.81 feet and 6.3 feet

variance with z-table

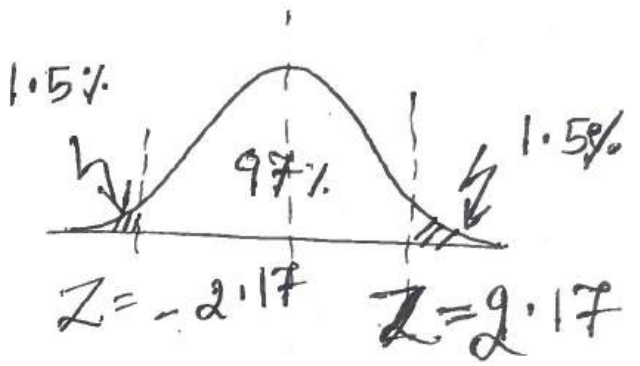
5) In a sample of 400 selected at random, a sample mean of 50 was obtained. Determine the confidence interval with a confidence level of 97% of the average popap population. The standard deviation is 2.

$m = 50 \qquad N = 400 \qquad \sigma = 2$

97% confidence

From (z-table)

z	A(z)
2.17	0.9850



∴ Confidence Interval

$$\left(m - 2.17 \times \frac{\sigma}{\sqrt{N}} , \; m + 2.17 \times \frac{\sigma}{\sqrt{N}} \right)$$

$$= \left(50 - \frac{2.17 \times 2}{\sqrt{400}} , \; 50 + \frac{2.17 \times 2}{\sqrt{400}} \right)$$

$$= (49.783 , \; 50.217)$$

(41)

① The average height of a random sample of 400 people from a city is 1.75m. It is known that the height of the population are random variables that follows normal distribution with a variance of 0.16m. Determine the interval of 95% confidence for the average height of the population

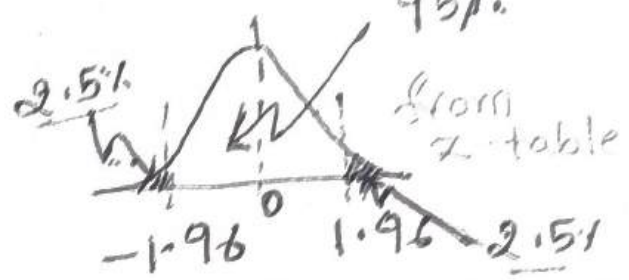
Solution:

$\sigma^2 = 0.16$ $N = n = 400$

$\sigma = \sqrt{0.16}$ $m = 1.75$

Std. deviation $\sigma = 0.4$

$$Z = \frac{m - \mu}{\frac{\sigma}{\sqrt{n}}}$$



For 95% confidence we have $P(-1.96 < Z < 1.96) = 0.95$

$$P\left\{ m - 1.96\frac{\sigma}{\sqrt{n}} < \mu < m + 1.96\frac{\sigma}{\sqrt{n}} \right\} = 0.95$$

↓ [from z-table]

$$P\left\{ 1.75 - \frac{1.96 \times 0.4}{\sqrt{400}} < \mu < 1.75 + \frac{1.96 \times 0.4}{\sqrt{400}} \right\}$$

z	A(z)
1.96	0.9750

← from z-table

$$P\left\{ 1.75 - \frac{1.96 \times 0.4}{\sqrt{400}} < \mu < 1.75 + \frac{1.96 \times 0.4}{\sqrt{400}} \right\} = 0.95$$

$$P\{ 1.7108 < \mu < 1.7892 \}$$

Interval of 95% confidence = (1.7108, 1.7892)

Page. No. 198 Table 9.3 - Research Methodology - Methods and Techniques q/c

Note :- Z-test and the test statistic

$$Z = \frac{X - \mu_{H0}}{\sigma_p / \sqrt{n}}$$

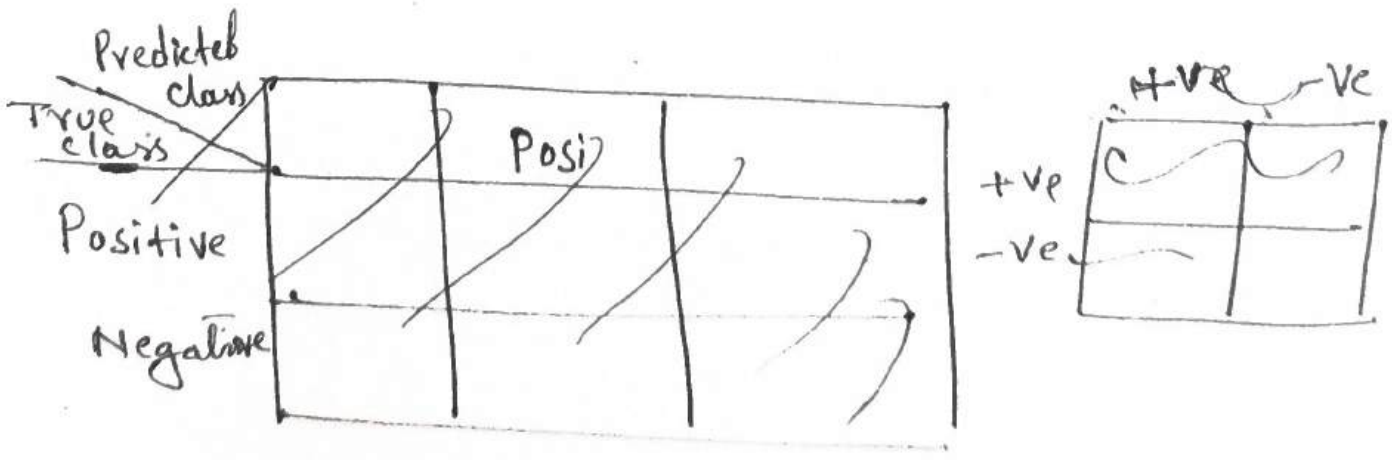
In case σ_p is not known, we use σ_s in its place

Calculating $\sigma_s = \sqrt{\frac{\Sigma (x_i - \bar{x})^2}{n-1}}$

Standard deviation Formula

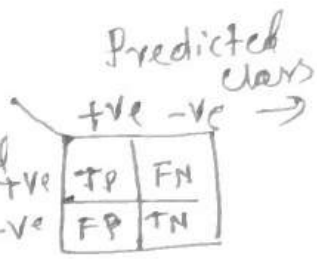
Population	Sample
$\sigma = \sqrt{\frac{\Sigma (X-\mu)^2}{N}}$	$S = \sqrt{\frac{\Sigma (X-\bar{x})^2}{n-1}}$
X - The value in the data distribution	X - The value in the data distribution
μ - The population mean	μ - The Sample mean
N - Total number of observations	n - Total number of observations

Ref: cuemath.com/data/standard-deviation



1) Find the True Positive, True Negative, False Positive and False Negative of the class setosa for the confusion matrix of the Iris dataset as shown in figure. (3) (4.9)

True class / Actual class	Predicted class: Setosa	Versicolar	Virginica
Setosa	16 (cell1)	0 (cell2)	0 (cell3)
Versicolor	0 (cell4)	17 (cell5)	1 (cell6)
Virginica	0 (cell7)	0 (cell8)	11 (cell9)



Solution: (i) Let us consider class setosa.

Let TP = True Positive TN = True Negative FP = False Positive

FN = False Negative, cell2 cell3

TP = 16 FN = 0 + 0 = 0 = Cell2 + cell3

False Positive = FP = cell4 cell7 0 + 0 = 0 = Cell4 + Cell7

TN = Cell5 + cell6 + cell8 + cell9 = 17 + 1 + 0 + 11 = 29

∴ TP = 16, FN = 0, FP = 0, TN = 29

(ii) 2) Repeat the above problem for class Versicolor

Solution: TP = 17

FN = cell4 + cell6 = 0 + 1 = 1

FP = Cell2 + cell8 = 0 + 0 = 0

TN = Cell1 + cell3 + cell7 + cell9 = 16 + 0 + 0 + 11 = 27

TP = 17 FN = 1 FP = 0 TN = 27

3) Repeat the above problem for class Virginica

Solution: TP = 11 FN = cell7 + cell8 = 0 + 0 = 0

FP = cell3 + cell6 = 0 + 1 = 1

TN = Cell1 + cell2 + cell4 + Cell5 = 16 + 0 + 0 + 17 = 33

2) The confusion matrix for four classes is shown in the figure. Find true positive (TP), true negative (TN), false positive (FP) and false negative (FN). for Class-1

Solution:

predicted class \ Expected class	1	2	3	4
1	52	3	7	2
2	2	28	2	0
3	5	2	25	12
4	1	1	9	40

Let us write the confusion matrix as follows.

Expected class \ Predicted class	1	2	3	4
1	52	2	5	1
2	3	28	2	1
3	7	2	25	9
4	2	0	12	40

⇒

TP	FN
52	8
12	119
FP	TN

For class-1:

TP = 52 FN = 8 FP = 12 TN = 119

3) Repeat the programmador above problem for class 2.

Solution:

	1	2	3	4
1	52	2	5	1
2	3	28	2	1
3	7	2	25	9
4	2	0	12	40

For class 2:

$TP = 28$ $\quad FN = 3+2+1 = 6$

$FP = 2+2+0 = 4$

$TN = 52+7+2+5+25+12+1+9+40$

$TN = 153$

4) Confusion Matrix

Actual \	Setosa	Versicolor	Virginica
Setosa	21	2	0
Versicolor	12	4	12
Virginica	5	3	16

predicted

For class Setosa:

	Setosa	Ver	Vir
Setosa	tp 21	fn 2	fn 0
Ver	fp 12	tn 4	tn 12
Vir	fp 5	tn 3	tn 16

TP = 21

TN = 4+12+3+16 = 35

FP = 12+5 = 17

FN = 2+0 = 2

For class Vericolor

	Setosa	Versicolor	Virginica
Setosa	tn 21	fp 2	tn 0
Versicolor	fn 12	tp 4	fn 12
Virginica	tn 5	fp 3	tn 16

TP = 4

TN = 21+0+5+16 = 42

FN = 12+12 = 24

FP = 2+3 = 5

For class Virginica

	Setosa	Versicolor	Virginica
Setosa	tn 21	tn 2	fp 0
Versicolor	tn 12	tn 4	fp 12
Virginica	fn 5	fn 3	tp 16

TP = 16

TN = 21+2+12+4 = 39

FP = 0+12 = 12

FN = 5+3 = 8

① A machine dispenses 80mL of fluid in a bottle on average. An employer studies 40 samples and finds out the sample mean to be 78 mL. Assume standard deviation to be 2.5. Write down the Null and Alternate hypothesis. At a 95% confidence level, is there enough evidence to support the idea that machine is not working properly?

Answer. ∴ Null Hypothesis: $H_0 : \mu_0 = 80$

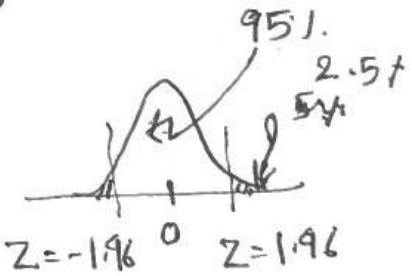
Alternative: $H_a : \mu \neq 80$

Sample mean $m = 78$

Sample size $N = 40$ $\quad \sigma = 2.5$

$$P\left\{ m - \frac{Z_{\alpha/2} \cdot \sigma}{\sqrt{N}} < \mu < m + \frac{Z_{\alpha/2}\,\sigma}{\sqrt{N}} \right\} = 0.95$$

95% confidence $\Rightarrow Z_{\alpha/2} = 1.96$



$$P\left\{ 78 - \frac{1.96 \times 2.5}{\sqrt{40}} < \mu < 78 + \frac{1.96 \times 2.5}{\sqrt{40}} \right\}$$

$$P\left\{ 77.2252 < \mu < 78.7747 \right\} = 0.95$$

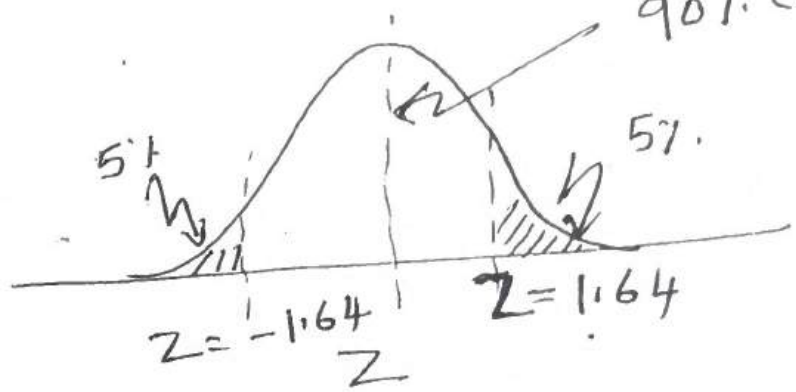
confidence interval (77.2252, 78.7747)

$H_0 : \mu = 80$ does not lie in the 95% confidence interval

∴ There is enough evidence to reject the null hypothesis H_0 at 95% confidence level

2) A machine dispenses 80 mL of fluid in a bottle on average. An employer studies 40 samples and finds the sample mean to be 78 mL. Assume $\sigma = 2.5$. Design a hypothesis test such that the significance level is 0.05

Solution:



from Z-table

$$P\left\{m > Z \times \frac{\sigma}{\sqrt{N}} < m < Z \times \frac{\sigma}{\sqrt{N}}\right\} = 0.90$$

$$m - Z \times \frac{\sigma}{\sqrt{N}} = \frac{78}{80} - \frac{1.64 \times 2.5}{\sqrt{40}} = 77.35$$

$$m + Z \times \frac{\sigma}{\sqrt{N}} = 78 + \frac{1.64 \times 2.5}{\sqrt{40}} = \mu, \sigma 2$$

$H_0 : \mu = 80$

$H_a : \mu \neq 80$

Population $\sim N(80, (2.5)^2)$

Sample mean, $m \sim N\left(80, \frac{2.5^2}{\sqrt{40}}\right)$

$m = 78$

Test: Reject H_0 if $m < c$

Significance level $= P\{\text{Reject } H_0 \mid H_0 \text{ is true}\}$

$= P\{m < c \mid \mu = 80\}$

$$P\{ N(0,1) < \frac{(C-80)\sqrt{40}}{\sigma} \}$$

$$P\{ N(\mu, \frac{\sigma^2}{N}) < C \}$$

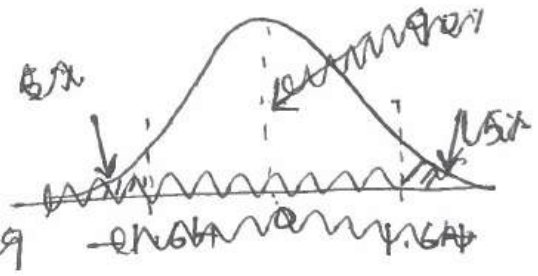
$$P\{ N(0,1) < \frac{(C-\mu)\sqrt{N}}{\sigma} \}$$

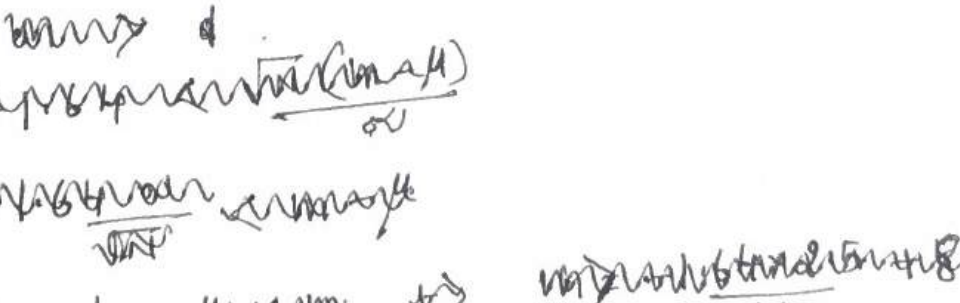
$$P\{ N(0,1) < \frac{(C-80)\sqrt{40}}{2.5} \} = 0.05$$

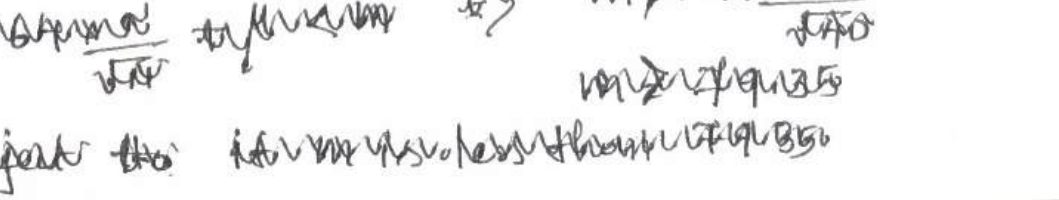
$$\frac{(C-80)\sqrt{40}}{2.5} = -1.64$$

$C = 79.35$ Reject H_0 if $m < 79.35$

Alternate Solution :







Unit-1 Numericals Contd..

1) Consider the set of inputs given for a Bootstrap experiment

$S = \{0.1, 0.7, 11.1, 7.5, 13.5, 7.79, 4.5, 10.01, 19.01, 11.45, 6.75, 8.87\}$

Samples are drawn after eight random picks are

$S' = \{0.7, 7.5, 0.7, 6.75, 10.01, 6.75, 11.1, 13.5\}$

a) Identify training and validation samples

b) After five picks identify out of box samples.

Solution:

a) $S = \{0.1, 0.7, 11.1, 7.5, 13.5, 7.79, 4.5, 10.01, 19.01, 11.45, 6.75, 8.87\}$

Training Samples = $S' = \{0.7, 7.5, 0.7, 6.75, 10.01, 6.75, 11.1, 13.5\}$

Validation Samples = $S^* = \{0.1, 7.79, 4.5, 19.01, 11.45, 8.87\}$

Out of box Samples

b) After five picks

Training Samples = $S' = \{0.7, 7.5, 0.7, 6.75, 10.01\}$

Validation / out of box samples = $S^* = \{0.1, 11.1, 13.5, 7.79, 4.5, 19.01, 11.45, 8.87\}$

2) Consider the confusion matrix given a two class classifier. Identify the missing data if the accuracy of classifier is 70% and the total number of samples are 1000

True class

Predicted class	A	B
A	345	76
B	x_2	x_1

Solution :-

Predicted class

True class	A	B
A	345 (tp)	x_2 (fn)
B	76 (fp)	x_1 (tn)

$$\text{Accuracy} = \frac{tp + tn}{N}$$

$$70\% = \frac{345 + x_1}{1000}$$

$$\frac{70}{100} = \frac{345 + x_1}{1000}$$

$$\frac{70 \times 1000}{100} - 345 = x_1$$

$$x_1 = 355$$

$$\boxed{TN = 355}$$

$$\text{Error} = 1 - \text{Accuracy}$$

$$= 1 - \frac{70}{100} = \frac{30}{100}$$

$$\frac{30}{100} = \frac{fn + fp}{1000}$$

$$\frac{30}{1000} \times 1000 = x_2 + 76$$

$$\frac{30}{100} \times 1000 - 76 = x_2$$

$$\boxed{x_2 = 224}$$

(3) Consider a three class ~~classifier~~ confusion matrix given. Calculate the total number of samples considerd for the experiment and the different performance metrics.

True class

predicted class	A	B	C
A	19	08	12
B	7	27	11
C	00	09	15

Solution: The given confusion matrix can be written as

Predicted class

True class	A	B	C
A	tp 19	fn 7	fn 00
B	fp 08	tn 27	tn 09
C	fp 12	tn 11	tn 15

predicted

True class	A	B
A	tp	fn
B	fp	tn

Let tp, tn, fp, fn for class A

Total number of samples = 19+7+8+27+9 +12+11+15 = 108

tp = 19 = 19

fn = 7+00 = 07

fp = 08+12 = 20

tn = 27+9+11+15 = 62

Accuracy for class A = $\frac{19+62}{19+7+20+62} = \frac{81}{108}$ =

(a) Accuracy = $\frac{19+27+15}{108} = 0.565$

$= 56.5\%$

(b) Error = $1 - \text{Accuracy} = 1 - 0.564 = 43.5$

(c)

predicted class

True class	A	B	C
A	19 tp	7 fn	00 fn
B	08 fp	27 tn	09 tn
C	12 fp	11 tn	15 tn

for A

	A	B	C
fn	7	17	23
tn	62	46	61
fp	20	18	09
tp	19	27	15
	108	108	108

Predicted class for B

True class	A	B	C
A	19 tn	7 fp	00 tn
B	08 tn	27 tp	09 fn
C	12 tn	11 fp	15 tn

for C

	A	B	C
A	19 tn	7 tn	00 fp
B	08 tn	27 tn	09 tp
C	12 fn	11 fn	15 tp

(d) Accuracy = $\frac{19+27+15}{108}$

c) tp-rate
Recall
Sensitivity $= \frac{tp}{tp+fn} = \frac{1}{3}\left[\frac{19}{19+7} + \frac{27}{27+17} + \frac{15}{15+23}\right]$

$= \frac{1}{3}\left[\frac{1}{3}\left[0.7307 + 0.6136 + 0.395\right]\right]$

$= 0.5797 = 57.97\%$

d) $\text{fp-rate} = \frac{fp}{fp+tn} = \frac{1}{3}\left[\frac{20}{20+62} + \frac{18}{18+46} + \frac{09}{9+61}\right]$

$= \frac{1}{3}\left[0.2439 + 0.28125 + 0.1285\right] = 0.2178$

$= 21.78\%$

c) $\text{precision} = \frac{tp}{tp+fp} = \frac{1}{3}\left[\frac{19}{19+20} + \frac{27}{27+18} + \frac{15}{15+9}\right]$

$= \frac{1}{3}\left[0.4871 + 0.6 + 0.625\right] = 0.5707$

$= 57.07\%$

f) $\text{Specificity} = 1 - \text{fp-rate} = 1 - 0.2178$

$= 78.22\%$

$= \frac{tn}{tn+fp} = \frac{1}{3}\left[\frac{62}{62+20} + \frac{46}{46+18} + \frac{61}{61+9}\right]$

$= \frac{1}{3}\left[0.7560 + 0.7187 + 0.8714\right]$

$= 0.7820 = 78.2\%$

Accuracy $= \frac{1}{3}\left[\frac{19+62}{108} + \frac{27+46}{108} + \frac{15+61}{108}\right]$

$= \frac{1}{3}\left[0.75 + 0.6759 + 0.7037\right]$

$= 0.7098$

$= 0.2902$

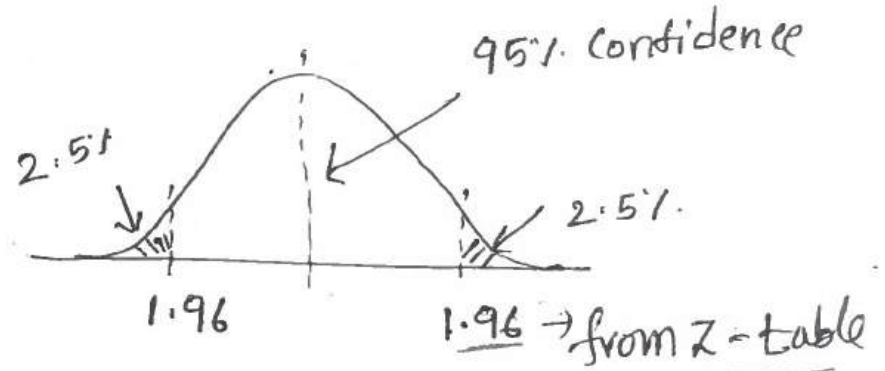
(4) A person want to rent an unfurnished one BHK apartment in Durham, NC next year. The mean monthly rent for a random sample of 60 apartments advertised on Craig's List (a website that list apartments for rent) is $1000. Assume a population standard deviation of $200. Construct a 95% confidence Interval.

Solution:

$m = \$1000$

$\sigma = \$200$

$N = 60$



$$\text{Confidence Interval} = \left\{ m - Z \times \frac{\sigma}{\sqrt{N}},\ m + Z \times \frac{\sigma}{\sqrt{N}} \right\}$$

$$= \left\{ 1000 - 1.96 \times \frac{200}{\sqrt{60}},\ m + 1.96 \times \frac{200}{\sqrt{60}} \right\}$$

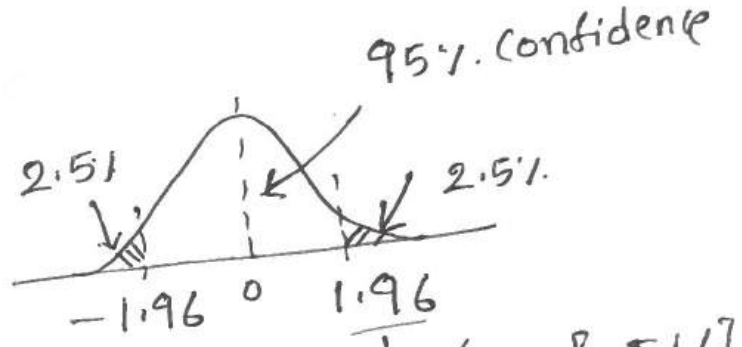
$$= \{ \$\ 949.39,\ \$\ 1050.606 \}$$

We are 95% confident that interval {$949.39, $1050.606} covers the true mean monthly rent of Durham apartments listed on Craig's List.

5) Duncan Jones kept careful records of the fuel efficiency of his car. After the first 100 times he filled up the tank, he found the mean was 23.4 miles per gallon (mpg) with a population standard deviation of 0.9 mpg. Compute 95% confidence interval for his mpg.

[1 gallon = 3.78 liters]

Solution:



↳ from [Z-Table]

$m = 23.4$

$Z = 1.96$

$\sigma = 0.9$

$N = 100$

Confidence interval $= \left(m - Z \times \frac{\sigma}{\sqrt{N}},\ m + Z \times \frac{\sigma}{\sqrt{N}}\right)$

$= \left(23.4 - \frac{1.96 \times 0.9}{\sqrt{100}},\ 23.4 + \frac{1.96 \times 0.9}{\sqrt{100}}\right)$

$= (23.22,\ 23.5764)$

We are 95% confidence that the interval (23.22 mpg, 23.5764 mpg) includes the true mean mpg of Duncan's automobile.

6) A candy machine makes chocolate bars that are on average 5gm. A worker claims that the machine after ~~maintain~~ maintenance no longer makes 5gm bars. Write the Null Hypothesis and Alternative Hypothesis

Solution:

$H_0 : \mu = 5gm$ Null Hypothesis

$H_a : \mu \neq 5gm$ Alternate Hypothesis

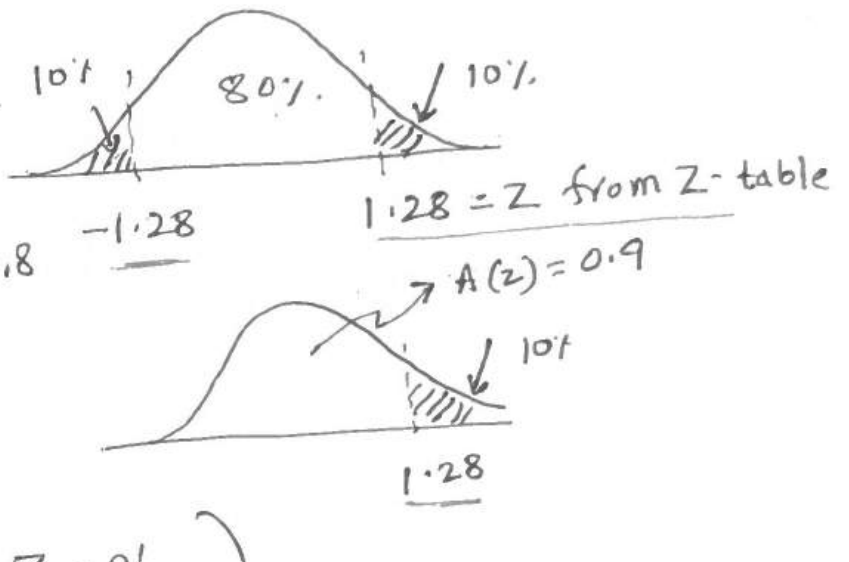
(1) Scores in an exam is Normally distributed with standard deviation = 5.6. A Random Sample of 40 scores on the exam has a mean of 32. Estimate the confidence Interval with 80% confidence, 90% confidence and 98% confidence

Solution:

(i) 80% Confidence

$m = 32$, $\sigma = 5.6$

$N = 40$



$$P\left\{m - Z \times \frac{\sigma}{\sqrt{N}} < \mu < m + Z \times \frac{\sigma}{\sqrt{N}}\right\} = 0.8$$

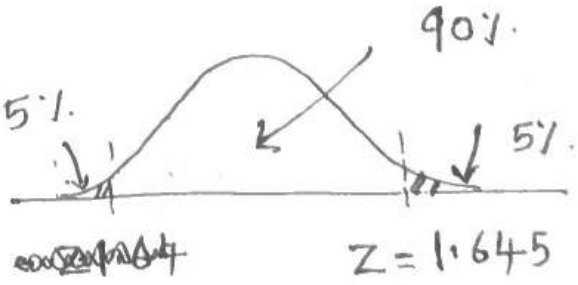
Confidence Interval

$$= \left[m - Z\frac{\sigma}{\sqrt{N}}, m + Z \times \frac{\sigma}{\sqrt{N}}\right)$$

$$= \left(32 - \frac{1.28 \times 5.6}{\sqrt{40}}, 32 + \frac{1.28 \times 5.6}{\sqrt{40}}\right)$$

$$= (30.8666, 33.13)$$

(ii) 90% Confidence



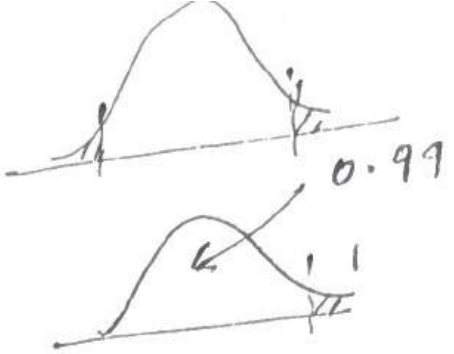
Confidence Interval

$$= \left(m - Z\frac{\sigma}{\sqrt{N}}, m + Z \times \frac{\sigma}{\sqrt{N}}\right)$$

$$= \left(32 - \frac{1.645 \times 5.6}{\sqrt{40}}, 32 + \frac{1.645 \times 5.6}{\sqrt{40}}\right)$$

$$= (30.54, 33.45)$$

c) 98% confidence



$z = 2.33$

$$\left(m - z \times \frac{\sigma}{\sqrt{N}},\ m + z \times \frac{\sigma}{\sqrt{N}}\right)$$

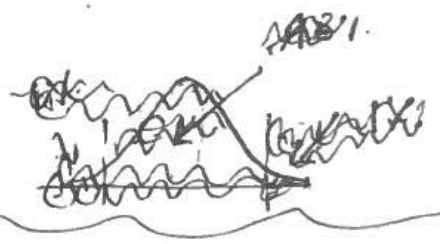
$$\left(32 - 2.33 \times \frac{5.6}{\sqrt{40}},\ 32 + 2.33 \times \frac{5.6}{\sqrt{40}}\right) = (29.936,\ 34.06)$$

$$= (29.94,\ 34.06)$$

As the confidence level increases, interval expands

~~b) 95% confidence~~

~~z = 1.96~~



2) It is thought that average life span of a car battery is $\geq$ 2 year. An engineer believes that the value is less using 30 samples, the engineer finds out that average life span is 1.8 years. At 99% confidence level, is there enough evidence to discard the null hypothesis. Assume standard deviation = 0.15.

Solution:

$H_0: \mu \geq 2$ [Null Hypothesis]

$H_a: \mu < 2$ [Alternative Hypothesis]

$N = 30 \quad m = 1.8 \quad \sigma = 0.15$

This will be 1-tail test



$$\therefore \text{Confidence Interval} = \left(-\infty,\ m + Z_{\alpha}\frac{\sigma}{\sqrt{n}}\right)$$

$$= \left(-\infty,\ 1.8 + \frac{2.33 \times 0.15}{\sqrt{30}}\right) = (-\infty,\ 1.8638)$$

Inference: The engineer establishes with 99% confidence level that the average life span of car battery lies in the range $(-\infty, 1.8638)$.

Therefore there is enough evidence to discard the null hypothesis $H_0: \mu \geq 2$

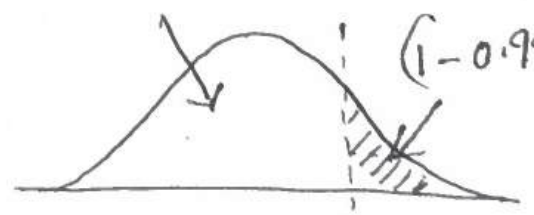
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3) The average test score for an entire school is 75 with $\sigma = 10$, what is the probability that a random sample of 30 students scored above 80?

Solution: $N = 30$, $m = 80$, $\sigma = 10$, $\mu = 75$

$$Z = \frac{m - \mu}{\frac{\sigma}{\sqrt{N}}} = \frac{80 - 75}{\frac{10}{\sqrt{30}}} = 2.738$$

$0.9968 \approx 0.9969$



∴ The Probability that a random sample of 30 students scored above 80 $= (1 - 0.9969) = 0.0031$

4) Consider a population with mean μ and Variance 1. The null and alternative hypothesis are $H_0: \mu = -1$

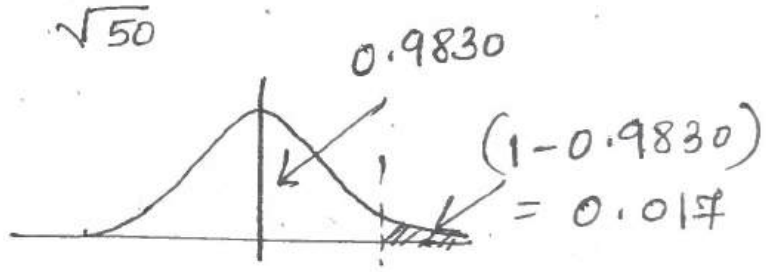
$H_a: \mu = 1$.

Sample mean on 50 samples is computed. Suppose the test is : Reject Ho if the Sample mean is greater than -0.7. Calculate the Significance level and power of the test.

Solution: $N = 50$ $m = -0.7$ $\mu = -1$, $\sigma = 1$

Test Reject Ho if $m > -0.7$ $H_0: \mu = -1$

$$Z = \frac{m - \mu}{\frac{\sigma}{\sqrt{N}}} = \frac{-0.7 + 1}{\frac{1}{\sqrt{50}}} = 2.12$$



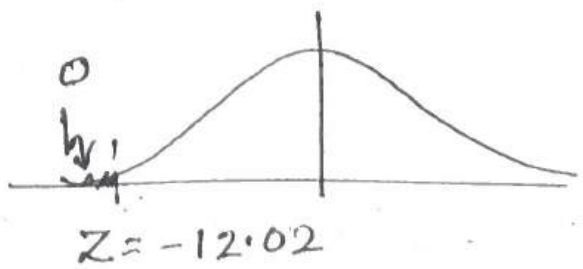
Significance level = 0.017

Probability to Reject Ho if there sample mean is greater than mean? Here $H_a: \mu = 1$

β = Probability { Fail to reject Ho | when Ha is true }

$\mu = 1$ $$Z = \frac{m - \mu}{\frac{\sigma}{\sqrt{N}}} = \frac{-0.7 - 1}{\frac{1}{\sqrt{50}}} = -12.02$$

$\beta = 0$



Power of the test $= 1 - \beta = 1$

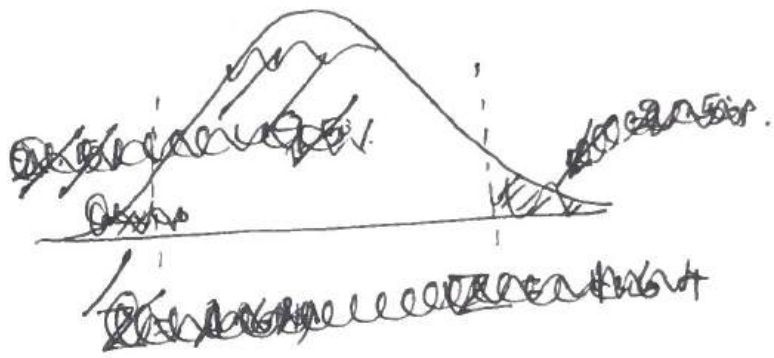
5) A machine dispenses 80ml of fluid in a bottle on average. An employer studies 40 samples and finds the sample mean to be 78 ml. Assume $\sigma = 2.5$. Design a hypothesis test such that the significance level is 0.05.

Solution: $\mu = 80$ $\sigma = 2.5$ $N = 40$

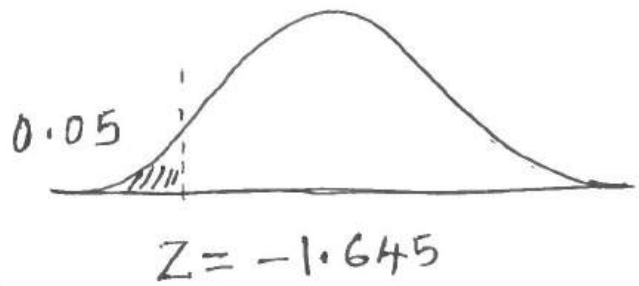
Significance level = 0.05

Null Hypothesis $H_0: \mu = 80$

Alternative Hypothesis $H_a: \mu \neq 80$



$$\frac{m - \mu}{\frac{\sigma}{\sqrt{N}}} = Z$$



$Z = -1.645$

$$\frac{m - 80}{\frac{2.5}{\sqrt{40}}} = -1.645$$

$$m - 80 = (-1.645)\frac{2.5}{\sqrt{40}}$$

$$m = 80 - \frac{1.645 \times 2.5}{\sqrt{40}}$$

$$m = 79.349$$

Test designed: Reject H_0 if $m < 79.349$

Since the sample mean $= m = 78$, null hypothesis H_0 is rejected with significance level 0.05

1) Given a machine learning model to classify calls as spam and non-spam, the following confusion matrix is generated. Find Accuracy, Precision, Recall, Specificity, error, fp-rate.

		Predicted: Spam	Predicted: non-spam	
Actual	Spam	118	12	130
	non-spam	47	15	62
		165	27	192

Acc = 69.2%

Pre = 71.5%

Re = 90.7%

Spe = 24.1%

error = 30.7%

fp-rate = 75.8%

2) Given a ML model to test the collected samples for diabetic (positive) and non-diabetic -(negative) classes, having the following confusion matrix. Find Accuracy, Precision, recall, specificity, sensitivity, tp-rate, fp-rate, error

Predicted \ Actual	Dia	Non-Dia
Dia	238	350
Non-Dia	150	100

Acc = 40.3%

Pre = 40.4%

Rec = 61.3%

Spe = 22.22%

fp-rate = 77.77%

error = 59.6%

3) An ML model generated the following confusion matrix. Find Error, Accuracy, Precision, Recall, Specificity, tp-rate, fp-rate.

		Predicted: Yes	Predicted: No
Reality	Yes	50	5
	No	10	35

N=100

4)

Actual \ Predicted	0	1	2
0	40	8	2
1	7	34	9
2	2	3	45

		class 0	1	2
Acc	79.3%	87.3%	82%	89.33%
Pre		81.6%	75.5%	80.3%
Re		80%	68%	90%
Spe		91%	89%	89.9%
error		12.66%	18%	10.67%
fp-rate		9%	11.1%	11%

The accuracies calculated are one-vs-rest accuracies. It is advantageous only we requires to measure how well model separates class i from all other classes combined.

Per-class accuracy = Recall.

Q Find the mean value of $x = [2, 4, 6, 8, 10]$ using bootstrapping method with sample replacement size of 5

Ans:-

Sample	Bootstrap Sample	mean
B_1	[2, 4, 6, 6, 10]	5.6
B_2	[4, 4, 6, 8, 10]	6.4
B_3	[2, 2, 6, 8, 10]	5.6
B_4	[4, 6, 6, 8, 10]	6.8
B_5	[2, 4, 6, 8, 10]	6

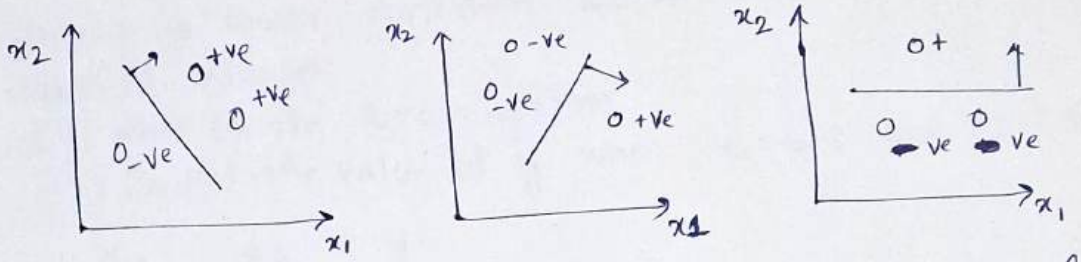
Bootstrap mean:-

$\hat{\mu} = 6.08$

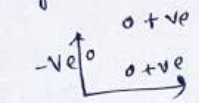
$$\text{Std. error} = \sqrt{\frac{\sum (m_i - \hat{\mu})^2}{B}} = \sqrt{\frac{1.088}{5}} = 0.4664$$

(1) show that VC dimension of a line is 3

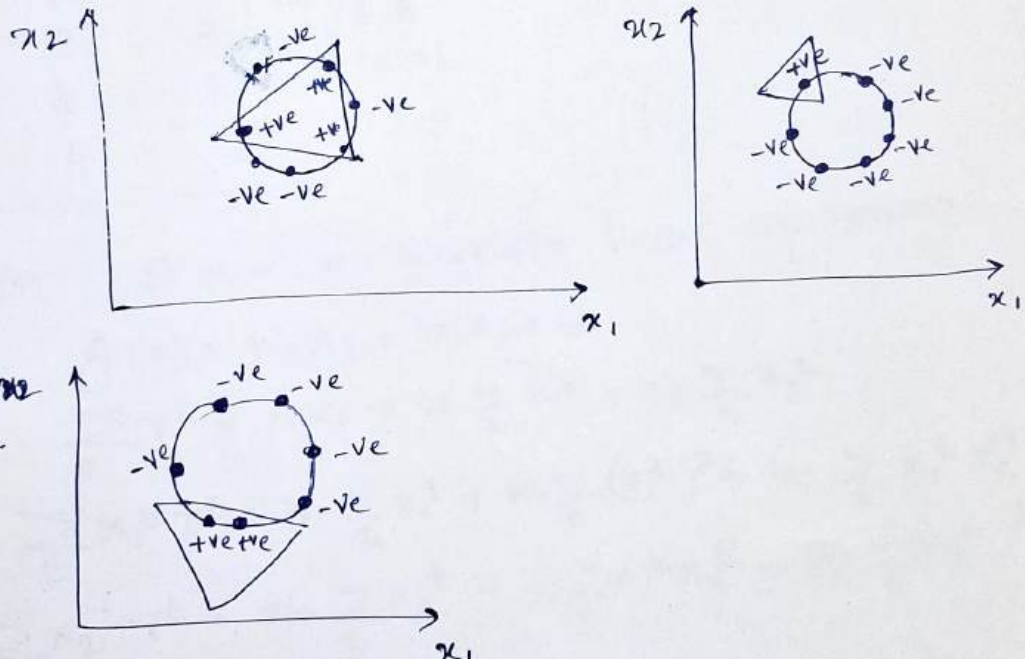
Solution :-



All possible labelings of three points can be Seperated using a line.

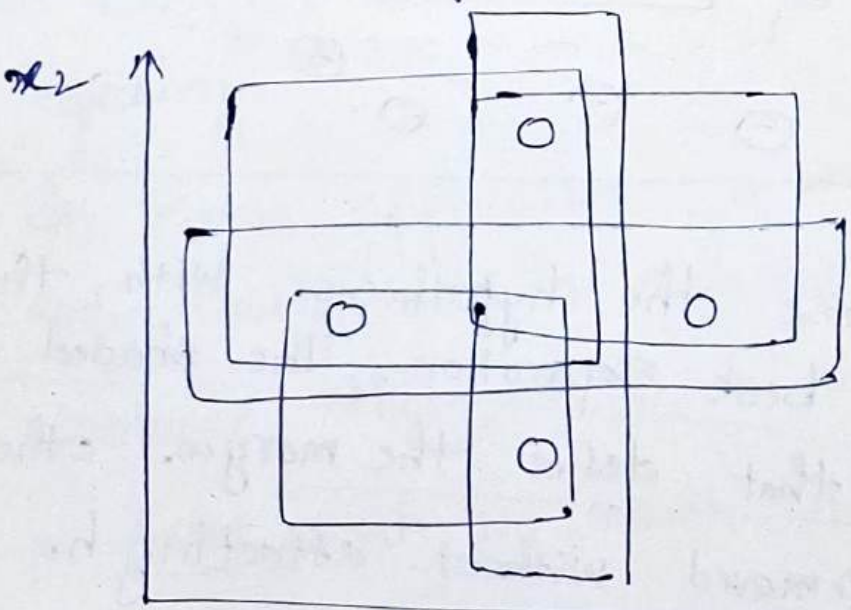


(2) show that the VC dimension of the triangle hypothesis Class is 7 in two dimensions

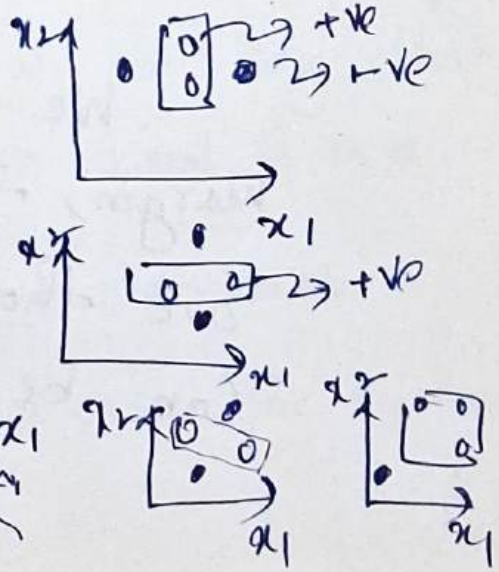


These 7 points can be Seperated

measures the



It is enough that we find four points that can be shattered.



$VC(H)$, when H is the hypothesis class of axis-aligned rectangles in two dimensions, is four.

Correct

2) For the following data fit a regression model

$g(x) = w_2 x^2 + w_1 x + w_0$

x	1	2	3
r^t	6	11	18

Answer :

$g(x^t) = r^t = x^2 + 2x + 3$

$$A = \begin{bmatrix} N & \sum_t x^t & \sum_t (x^t)^2 \\ \sum_t x^t & \sum_t (x^t)^2 & \sum_t (x^t)^3 \\ \sum_t (x^t)^2 & \sum_t (x^t)^3 & \sum_t (x^t)^4 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 6 & 14 \\ 6 & 14 & 36 \\ 14 & 36 & 98 \end{bmatrix} \quad w = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix} \quad y = \begin{bmatrix} \sum_t r^t \\ \sum_t r^t x^t \\ \sum_t r^t (x^t)^2 \end{bmatrix}$$

$$y = \begin{bmatrix} 35 \\ 82 \\ 212 \end{bmatrix}$$

$$Aw = y$$

$$w = A^{-1} y = \begin{bmatrix} 3 & 6 & 14 \\ 6 & 14 & 36 \\ 14 & 36 & 98 \end{bmatrix}^{-1} \begin{bmatrix} 35 \\ 82 \\ 212 \end{bmatrix}$$

$$w = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix}$$

$w_2 = 1 \quad w_1 = 2 \quad w_0 = 3$

(1)

hours of study	P/F
1	F
2	F
3	F
4	F
5	P
6	P
7	P
8	F → outlier
9	P
10	P

(2)

	x	target, t
Training set	1	5.1
	2	10.8
	3	21.1
	1.5	7.05
	2.5	15.8
	3.5	26.5
Testing set	1.8	9.5
	2.2	12.5

N = 8
univariate problem

Split into training and testing set (20%)
Write the Loss function assuming a linear dependency of target on the feature.
Train the model. Report training error.
Report testing error too.

$$\mathcal{L} = \frac{1}{6} \sum_{i=1}^{6} \left(t_i - (w_0 + w_1 x_i) \right)^2$$

Training set

x	t
1	5.1
2	10.8
3	21.1
1.5	7.05
2.5	15.8
3.5	26.5

$$\begin{bmatrix} N & \sum_{i=1}^{N} x_i \\ \sum_{i=1}^{N} x_i & \sum_{i=1}^{N} x_i^2 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{N} t_i \\ \sum_{i=1}^{N} x_i t_i \end{bmatrix}$$

$$\begin{bmatrix} 6 & 13.5 \\ 13.5 & \end{bmatrix}$$

Q1

(3) Minimize $f(x) = (x-5)^2$ using gradient descent algorithm. Assume initial value of $x = 0$ and learning rate $= 0.1$. (2)

$$x_{k+1} = x_k - \alpha \frac{\partial f}{\partial x}$$

$x_0 = 0$, $x_1 = x_0 - 0.1 \times \frac{\partial}{\partial x}(x-5)^2 \Big|_{x=0}$

$= x_0 - 0.1 \times \frac{\partial}{\partial x}(x^2 - 10x + 25) \Big|_{x=0}$

$= x_0 - 0.1(2x - 10) \Big|_{x=0}$

$= 0 - 0.1(-10) = 1$

$x_2 = x_1 - 0.1 \times \frac{\partial}{\partial x}(x^2 - 10x + 25) \Big|_{x=1}$

$= 1 - 0.1 \times (2x - 10) \Big|_{x=1}$

$= 1 - 0.1 \times (-8)$

$= 1.8$

$x_3 = x_2 - 0.1(2x - 10) \Big|_{x=1.8}$

$= 1.8 - 0.1(3.6 - 10)$

$= 1.8 - 0.1(-6.4)$

$= 2.44$

$x_4 = x_3 - 0.1(2x - 10) \Big|_{x=2.44}$

$= 2.44 - 0.1(4.88 - 10)$

$= 2.952$

$x_5 = x_4 - 0.1(2x - 10) \Big|_{x=2.952}$

$= 2.952 - 0.1(-4.096)$

$= 3.3616$

$$x_6 = 3.3616 - 0.1(6.7232 - 10)$$

$$= 3.6892$$

$$x_7 = x_6 - 0.1(2x - 10)\big|_{x = 3.6892}$$

$$= 3.6892 - 0.1(-2.6216)$$

$$= 3.95136$$

$x_8 = 4.1611$

$x_9 = 4.329$

$x_{10} = 4.463$

$x_{11} = 4.5704$

$x_{12} = 4.65632$

$x_{13} = 4.725056$

We stop further iterations when change in x between successive iterations is very small.

Repeat assuming $\alpha = 0.5$

$x_0 = 0$

$$x_1 = x_0 - \alpha \frac{\partial f}{\partial x}\Big|_{x = x_0}$$

$$= 0 - 0.5(-10)$$

$$= 5$$

$$x_2 = x_1 - 0.5(2x - 10)\big|_{x=5}$$

$$= 5 - 0.5 \times 0$$

$$= 5$$

$x_2 = x_1$, stop the algorithm.

Repeat assuming $\alpha = 0.7$

$x_0 = 0$

$$x_1 = 0 - 0.7(2x - 10)\big|_{x=0}$$

$$= 0 - 0.7(-10) = 7$$

$$x_2 = 7 - 0.7(2x - 10)\big|_{x=7} = 4.2$$

$$x_3 = 4.2 - 0.7(2x - 10)\big|_{x=4.2}$$

$$= 5.32$$

$$x_4 = 5.32 - 0.7(2x - 10)\big|_{x=5.32}$$

$$= 4.87$$

$$x_5 = 5.05$$

$$x_6 = 4.97$$

$$x_7 = 5.008$$

$$x_8 = 4.996$$

$$x_9 = 5.001$$

Observe that the value of x gets closer and closer to the minima, back and forth.

If $|x_{k+1} - x_k| < 0.001$

stop the algorithm.

(4) Minimize $\mathcal{L} = (x-1)^2 + (y-2)^2$

Assume $\begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\alpha = 0.1$.

$$x_{k+1} = x_k - \alpha \frac{\partial L}{\partial x}\Big|_{x=x_k, y=y_k}$$

$$y_{k+1} = y_k - \alpha \frac{\partial L}{\partial y}\Big|_{x=x_k, y=y_k}$$

$$L = x^2 + 1 - 2x + y^2 + 4 - 2y$$

$$x_6 = 3.3616 - 0.1(6.7232 - 10)$$

$$= 3.6892$$

$$x_7 = x_6 - 0.1(2x - 10)\big|_{x = 3.6892}$$

$$= 3.6892 - 0.1(-2.6216)$$

$$= 3.95136$$

$x_8 = 4.1611$

$x_9 = 4.329$

$x_{10} = 4.463$

$x_{11} = 4.5704$

$x_{12} = 4.65632$

$x_{13} = 4.725056$

We stop further iterations when change in x between successive iterations is very small.

Repeat assuming $\alpha = 0.5$

$x_0 = 0$

$$x_1 = x_0 - \alpha \frac{\partial f}{\partial x}\Big|_{x = x_0}$$

$$= 0 - 0.5(-10)$$

$$= 5$$

$$x_2 = x_1 - 0.5(2x - 10)\big|_{x=5}$$

$$= 5 - 0.5 \times 0$$

$$= 5$$

$x_2 = x_1$, stop the algorithm.

Repeat assuming $\alpha = 0.7$

$x_0 = 0$

$$x_1 = 0 - 0.7(2x - 10)\big|_{x=0}$$

$$= 0 - 0.7(-10) = 7$$

$$x_2 = 7 - 0.7(2x - 10)\big|_{x=7} = 4.2$$

$$x_1 = 0 - 0.1(2x-2)\big|_{x=0}$$

$$= 0 - 0.1(-2) = 0.2$$

$$y_1 = 0 - 0.1(2y-4)\big|_{y=0} = -0.1(-4) = 0.4$$

$$x_2 = 0.2 - 0.1(2x-2)\big|_{x=0.2}$$

$$= 0.2 + 0.16 = 0.36$$

$$y_2 = 0.4 - 0.1(2y-4)\big|_{y=0.4}$$

$$= 0.4 + 0.32 = 0.72$$

$$x_3 = 0.36 - 0.1(2x-2)\big|_{x=0.36}$$

$$= 0.36 + 0.128 = 0.488$$

$$y_3 = 0.72 - 0.1(2y-4)\big|_{y=0.72}$$

$$= 0.72 + 0.328 = 0.976$$

$$x_4 = 0.59$$

$$y_4 = 1.18$$

(5)

x	t (target)
x_1	t_1
x_2	t_2
x_3	t_3
x_N	t_N

regression problem

$h(x) = w_0 + w_1 x$

$$E = \frac{1}{N} \sum_{i=1}^{N} (w_0 + w_1 x_i - t_i)^2$$

$$\frac{\partial E}{\partial w_0} = 0 , \quad \frac{\partial E}{\partial w_1} = 0$$

Therefore,

$$N w_0 + w_1 \sum_{i=1}^{N} x_i = \sum_{i=1}^{N} t_i \quad \cdots (1)$$

$$w_0 \sum_{i=1}^{N} x_i + w_1 \sum_{i=1}^{N} x_i^2 = \sum_{i=1}^{N} x_i t_i \quad \cdots (2)$$

$$\begin{bmatrix} N & \sum_{i=1}^{N} x_i \\ \sum_{i=1}^{N} x_i & \sum_{i=1}^{N} x_i^2 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{N} t_i \\ \sum_{i=1}^{N} x_i t_i \end{bmatrix}$$

$$\underset{\sim}{A} \quad \underline{w} \quad = \underline{b}$$

Or, $\underline{w} = \underset{\sim}{A}^{-1} \underline{b}$

(4)

x	t
1	6.1
2	7.8
3	10.1
4	12.5

$$\begin{bmatrix} N & \sum_{i=1}^{N} x_i \\ \sum_{i=1}^{N} x_i & \sum_{i=1}^{N} x_i^2 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{N} t_i \\ \sum_{i=1}^{N} x_i t_i \end{bmatrix}$$

Q1 $$\begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} = \begin{bmatrix} 36.5 \\ 102 \end{bmatrix}$$

Q1 $$\begin{bmatrix} w_0 \\ w_0 \end{bmatrix} = \begin{bmatrix} 1.5 & -0.5 \\ -0.5 & 0.2 \end{bmatrix} \begin{bmatrix} 36.5 \\ 102 \end{bmatrix} = \begin{bmatrix} 3.75 \\ 2.15 \end{bmatrix}$$

(6) Parametric Classification

Class conditional density is assumed to be Gaussian.

Salary (x_1)	Age (x_2)	Type of car (t)
90	35	I ✓
80	40	I ✓
60	40	II
30	40	II
62	30	I ✓
20	62	II
50	45	II

$N = 7$

$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$p(C_I) = \frac{3}{7}, \quad p(C_{II}) = \frac{4}{7}$

$$\hat{\underline{\mu}}_1 = \begin{bmatrix} \frac{90+80+62}{3} \\ \frac{35+40+30}{3} \end{bmatrix} = \begin{bmatrix} 77.33 \\ 35 \end{bmatrix}$$

$$\hat{\underline{\mu}}_2 = \begin{bmatrix} \frac{60+30+20+50}{4} \\ \frac{40+40+62+45}{4} \end{bmatrix} = \begin{bmatrix} 40 \\ 46.75 \end{bmatrix}$$

$$\hat{\underset{\sim}{\Sigma}}_1 = \begin{bmatrix} \frac{1}{3}\left[(90-77.33)^2 + (80-77.33)^2 + (62-77.33)^2\right] & \frac{1}{3}\left[(90-77.33)(35-35) + (80-77.33)(40-35) + (62-77.33)(30-35)\right] \\ & \frac{1}{3}\left[(35-35)^2 + (40-35)^2 + (30-35)^2\right] \end{bmatrix}$$

$$= \begin{bmatrix} 134.25 & 30 \\ 30 & 16.667 \end{bmatrix}$$

Similarly construct $\hat{\underset{\sim}{\Sigma}}_2 =$

$$\begin{bmatrix} \frac{1}{4}\left[(60-40)^2 + (30-40)^2 + (20-40)^2 + (50-40)^2\right] & \frac{1}{4}\left[(60-40)(40-46.75) + (30-40)(40-46.75) + (20-40)(62-46.75) + (50-40)(45-46.75)\right] \\ \frac{1}{4}\left[(40-46.75)^2 + (40-46.75)^2 + (62-46.75)^2 + (45-46.75)^2\right] & \end{bmatrix}$$

$$g_I(\underline{x}) = \ln\left\{p(\underline{x}|C_I)\, p(C_I)\right\}$$

$$g_{II}(\underline{x}) = \ln\left\{p(\underline{x}|C_{II})\, p(C_{II})\right\}$$

Decision Rule $\quad g_I(\underline{x}) \underset{II}{\overset{I}{\gtrless}} g_{II}(\underline{x})$

$$p(\underline{x}|C_I) = \frac{1}{(2\pi)^{d/2}\left|\underset{\sim}{\Sigma}_1\right|^{1/2}} e^{-\frac{1}{2}(\underline{x}-\underline{\mu}_1)^T \underset{\sim}{\Sigma}_1^{-1}(\underline{x}-\underline{\mu}_1)}$$

$$p(\underline{x}|C_{II}) = \frac{1}{(2\pi)^{d/2}\left|\underset{\sim}{\Sigma}_2\right|^{1/2}} e^{-\frac{1}{2}(\underline{x}-\underline{\mu}_2)^T \underset{\sim}{\Sigma}_2^{-1}(\underline{x}-\underline{\mu}_2)}$$

In this example, $d = 2$.

(7) Consider a binary classification problem with 2D feature vector. The class conditional densities are multivariate Gaussian.

$$C_I = \mathcal{N}\left(\begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}\right)$$

$$C_{II} = \mathcal{N}\left(\begin{bmatrix}3\\2\end{bmatrix}, \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}\right)$$

Assume a priori probabilities as 0.4 and 0.6

(a) Derive the discriminant functions.

(b) Determine the decision boundary.

(c) Classify the sample $\underline{x} = \begin{bmatrix}2.5\\1.2\end{bmatrix}$

(a) $g_I(\underline{x}) = \ln\{p(\underline{x}|C_I)\,p(C_I)\}$

$g_{II}(\underline{x}) = \ln\{p(\underline{x}|C_{II})\,p(C_{II})\}$

$g_I(\underline{x}) \underset{II}{\overset{I}{\gtrless}} g_{II}(\underline{x})$: decision rule

$d = 2$ here

$g_I(\underline{x})$

$= \ln\left\{p(\underline{x}|C_I)\,p(C_I)\right\}$

$= -\frac{d}{2}\ln(2\pi) - \frac{1}{2}\ln(|\underset{\sim}{\Sigma}_1|) - \frac{1}{2}(\underline{x}-\underline{\mu}_1)^T \underset{\sim}{\Sigma}_1^{-1}(\underline{x}-\underline{\mu}_1)$

$+ \ln(0.4)$

$= -\ln(2\pi) - \frac{1}{2}\ln(1) - \frac{1}{2}\begin{bmatrix}x_1-1 & x_2-1\end{bmatrix}\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}x_1-1\\x_2-1\end{bmatrix}$

$+ \ln(0.4)$

$$= -\ln(2\pi) - 0 - \frac{1}{2}\left\{(x_1-1)^2 + (x_2-1)^2\right\} + \ln(0.4) \quad (7)$$

$$= -\frac{x_1^2}{2} - \frac{x_2^2}{2} + x_1 + x_2 - 1.916$$

$$g_{II}(\underline{x}) = -\frac{d}{2}\ln(2\pi) - \frac{1}{2}\ln\left|\underset{\sim}{\Sigma}_2\right| - \frac{1}{2}(\underline{x}-\underline{\mu}_2)^T \underset{\sim}{\Sigma}_2^{-1}(\underline{x}-\underline{\mu}_2) + \ln(0.6)$$

$$= -\ln(2\pi) - \frac{1}{2}\ln(1) - \frac{1}{2}\left(\begin{bmatrix} x_1-3 & x_2-2 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x_1-3 \\ x_2-2 \end{bmatrix}\right) + \ln(0.6)$$

$$= -\ln(2\pi) - 0 - \frac{1}{2}\left((x_1-3)^2 + (x_2-2)^2\right) + \ln(0.6)$$

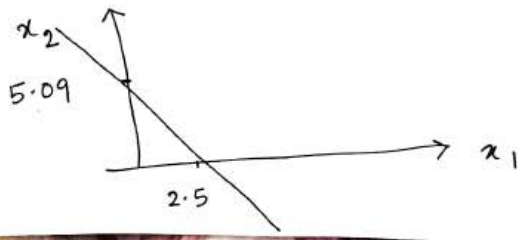
$$= -\frac{x_1^2}{2} - \frac{x_2^2}{2} + 3x_1 + 2x_2 - 7.0108$$

(b) Decision Boundary :

$$g_I(\underline{x}) = g_{II}(\underline{x})$$

$$-1.916 - \frac{x_1^2}{2} - \frac{x_2^2}{2} + x_1 + x_2 = -7.0108 - \frac{x_1^2}{2} - \frac{x_2^2}{2} + 3x_1 + 2x_2$$

or, $2x_1 + x_2 - 5.0948 = 0$



Decision boundary is a straight line

PES2UG24EC039

(c) $\underline{x} = \begin{bmatrix} 2.5 \\ 1.2 \end{bmatrix}$

Decision rule

$$g_I(\underline{x}) \underset{C_{II}}{\overset{C_I}{\gtrless}} g_{II}(\underline{x})$$

$$-1.916 - \frac{(2.5)^2}{2} - \frac{(1.2)^2}{2} + 2.5 + 1.2 \underset{C_{II}}{\overset{C_I}{\gtrless}}$$

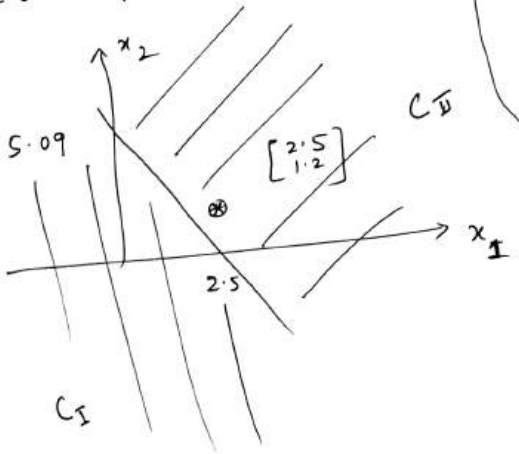
$$-7.0108 - \frac{(2.5)^2}{2} - \frac{(1.2)^2}{2} + 3(2.5) + 2(1.2)$$

or, $-2.06 \underset{C_{II}}{\overset{C_I}{\gtrless}} -0.99$

There $\underline{x}$ belongs to class II because

$-2.06 < -0.99$.

Since $|\Sigma_1| = |\Sigma_2|$ decision boundary is ast. line



(7) Assume $\underline{\mu}_2 = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$

and redo the calculations

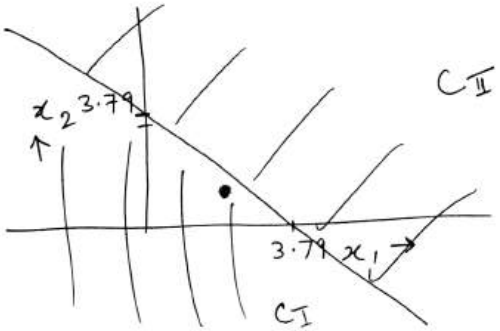
$$g_{II}(\underline{x}) = -9.5108 - \frac{x_1^2}{2} - \frac{x_2^2}{2} + 3x_1 + 3x_2$$

Decision boundary, $g_I(\underline{x}) = g_{II}(\underline{x})$

or, $-1.916 - \frac{x_1^2}{2} - \frac{x_2^2}{2} + x_1 + x_2$

$$= -9.5108 - \frac{x_1^2}{2} - \frac{x_2^2}{2} + 3x_1 + 3x_2$$

or, $x_1 + x_2 - 3.7974 = 0$



$$g_I(\underline{x})\Big|_{\underline{x} = \begin{bmatrix} 2.5 \\ 1.2 \end{bmatrix}} = -2.06$$

$$g_{II}(\underline{x})\Big|_{\underline{x} = \begin{bmatrix} 2.5 \\ 1.2 \end{bmatrix}} = -2.2558$$

$g_I(\underline{x}) > g_{II}(\underline{x})$

Therefore $\underline{x}$ belongs to C_I

(8) A random sample of 50 students has a mean score of 72. The population standard deviation is 8. Calculate 95% (double sided) confidence interval for population mean.

sample size, $N = 50$

sample mean $= m = 72$

$\sigma = 8$

confidence $= 0.95$

$$\frac{\sqrt{N}}{\sigma}(m-\mu) = Z \sim N(0,1)$$

$$\text{Prob}\ (-1.96 < Z < 1.96) = 0.95$$

Or, $\text{Prob}\left(\left(m - 1.96 \times \frac{\sigma}{\sqrt{N}}\right) < \mu < \left(m + 1.96 \times \frac{\sigma}{\sqrt{N}}\right)\right) = 0.95$

Or, $\text{Prob}\left(\left(72 - 1.96 \times \frac{8}{\sqrt{50}}\right) < \mu < \left(72 + 1.96 \times \frac{8}{\sqrt{50}}\right)\right) = 0.95$

Or, $\text{Prob}\ (69.7825 < \mu < 74.2174) = 0.95$

Double sided confidence Interval is

$(69.7825, 74.2174)$

(9) Repeat the above : calculate 95% single sided confidence interval for population mean.

(10) Average lifetime of a brand of (9) 100 batteries is 520 hours. Population std. dev. is 40 hours. Find 99% double sided confidence interval for the true mean lifetime.

$N = 100$

$m = 520$

$\sigma = 40$

Confidence $= 0.99$

Prob $(-2.58 < z < 2.58) = 0.99$

or, Prob $\left(m - \frac{2.58 \times \sigma}{\sqrt{N}} < \mu < m + \frac{2.58 \times \sigma}{\sqrt{N}}\right) = 0.99$

or, Prob $\left(\left(520 - \frac{2.58 \times 40}{\sqrt{100}}\right) < \mu < \left(520 + 2.58 \times \frac{40}{\sqrt{100}}\right)\right)$

$= 0.99$

or, Prob $(509.68 < \mu < 530.32) = 0.99$

Ans: double sided confidence interval is $(509.68, 530.32)$

(11) Population st. dev = 15. How large a sample is required so that a 95% upper confidence bound estimates μ within 4 unit ?

$$\frac{\sqrt{N}}{\sigma}(m-\mu) = Z$$

$\sigma = 15$

$N = ?$

confidence = 95% (upper)

$$\text{Prob}\left(Z < 1.64\right) = 0.95$$

or, $$\text{Prob}\left(\frac{\sqrt{N}}{\sigma}(m-\mu) < 1.64\right) = 0.95$$

or. $$\text{Prob}\left((m-\mu) < \frac{1.64\times\sigma}{\sqrt{N}}\right) = 0.95$$

or, $$\text{Prob}\left(-\mu < \left(\frac{1.64\sigma}{\sqrt{N}} - m\right)\right) = 0.95$$

or, $$\text{Prob}\left(\mu > \left(m - \frac{1.64\sigma}{\sqrt{N}}\right)\right) = 0.95$$

Error in estimate of μ is

$$\frac{1.64\sigma}{\sqrt{N}} = 4$$

$$\therefore \quad N = 37.8225$$

Ans. $\boxed{N = 38}$

(12) $m = \frac{\sum_{i=1}^{N} x_i}{N}$, $s^2 = \frac{\sum_{i=1}^{N} (x_i - m)^2}{N-1}$ (10)

If σ is not known

$$\mu = m \pm t_{\alpha/2, N-1} \left(\frac{s}{\sqrt{N}} \right)$$

If σ is known

$$\mu = m \pm Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{N}} \right)$$

No of samples = 16, sample mean is 40, Sample standard deviation is 4. Find the 95% confidence interval.

$N = 16$, $m = 40$, $s = 4$, $\alpha = (1 - 0.95)$

$= 0.05$

$\frac{\alpha}{2} = 0.025$

$$\mu = 40 \pm t_{0.025, 15} \left(\frac{4}{\sqrt{16}} \right)$$

$$= 40 \pm t_{0.025, 15}^{(1)}$$

~~$t_{0,0.25}$~~

$t_{0.025,15}^{(1)} = 2.131$

$$= 40 \pm 2.131$$

Confidence interval $(37.869, 42.131)$

(13) A manufacturer claims that the mean battery life is at least 10 hours. A sample of 49 batteries has a mean life of 9.5 hrs. Population std deviation = 1.4 hr. Test the hypothesis at 5% significance level. Use one-tailed test

$\mu = 10$

$m = 9.5$

$N = 49$

$\sigma = 1.4$

$\frac{\sqrt{N}}{\sigma}(m-\mu) = Z$

$Z \sim N(0,1)$

$H_0 : \mu \geq 10$

$H_1 : \mu < 10$

$\text{Prob}(Z < 1.64) = 0.95$

or, $\text{Prob}\left((m-\mu) < \frac{1.64\cdot\sigma}{\sqrt{N}}\right) = 0.95$

or, $\text{Prob}\left((-.5) < \left(\frac{1.64 \times 1.4}{7}\right)\right) = 0.95$

↑ violates ← +ve quantity

It means mean battery is not 10 hours.

Alternately

$(m-\mu)\frac{\sqrt{N}}{\sigma} = -2.5$

↑ scaled deviation $= -2.5$

allowed deviation < 1.64

Therefore H_0 fails

UE23EC352B -Machine Learning and Applications -UNIT – 2-Numericals

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Regression (Exercise)

Find the model g(x) to approximate the output using the following training samples with linear g(x).

{(0,7),(1,5),(2,4),(3,3),(4,3),(5,3),(6,2),(7,0)}

$$\mathcal{X} = \{\mathbf{x}^t, r^t\}_{t=1}^{N}$$

$$E(w_1, w_0|\mathcal{X}) = \frac{1}{N}\sum_{t=1}^{N}[r^t - (w_1 x^t + w_0)]^2$$

$$\begin{aligned} \sum_t r^t &= N w_0 + w_1 \sum_t x^t \\ \sum_t r^t x^t &= w_0 \sum_t x_t + w_1 \sum_t (x^t)^2 \end{aligned}$$

$$\begin{aligned} w_1 &= \frac{\sum_t x^t r^t - \overline{x}\,\overline{r}N}{\sum_t (x^t)^2 - N\overline{x}^2} \\ w_0 &= \overline{r} - w_1 \overline{x} \end{aligned}$$

$$g(\mathbf{x}) = w_1 x_1 + \cdots + w_d x_d + w_0 = \sum_{j=1}^{d} w_j x_j + w_0$$

$$\overline{x} = \sum_t x^t / N$$

$$\overline{r} = \sum_t r^t / N$$

- In parametric classification discriminant function is given by $g_i(x)=-(x-m_i)^2$ .If the classes given are C_1, C_2 and C_3, with mean $m_1 = 3$, $m_2 = -3$ and $m_3 = 4$, identify the class to which a given samples $x_1 = 5$ and $x_2 = -1$ should belong to. Assume the sample should be assigned to the class Ci

- In parametric classification discriminant function is given by $g_i(x)=-(x-m_i)^2$ .If the classes given are C_1, C_2 and C_3, with mean $m_1 = 3$, $m_2 = -3$ and $m_3 = 4$, identify the class to which a given samples $x_1= 5$ and $x_2 = -1$ should belong to. Assume the sample should be assigned to the class Ci if
- a.x1 belongs to class C1 and x2 to Class C1
- b.x1 belongs to class C3 and x2 to Class C2
- c.x1 belongs to class C2 and x2 to Class C3
- d.x1 belongs to class C2 and x2 to Class C1
-
-
- Ans: b
-

(2) In parametric classification discriminant function is given by $g_i(x) = -(x - m_i)^2$.
The sample x is assigned to the class with the nearest mean. Assume the number of classes K = 2 & mean $m_1 = -2$ and mean $m_2 = 2$
If the given sample $x' = 3$ find the class to which it belongs to

Solution:

x is assigned to class C_i with the nearest mean

- choose C_i if $|x - m_i| = \min_k |x - m_k|$

$m_1 = -2$ $|(x) - (-2)| = |3 + 2| = 5 \Rightarrow |3+2|$

$m_2 = +2$ $|(x) - (2)| = |-1| = 1 \Rightarrow |3-2|$

It belongs to class C_2

(3) Repeat the above problem for $x = -1$

Solution:

choose C_i if $|x - m_i| = \min_K |x - m_K|$

$m_1 = -2$ $\quad |x - m_1| = |-1 - (-2)|$

$= |-1 + 2| = 1$

$m_2 = 2$ $\quad |x - m_1| = |-1 - 2|$

$= 3$

It belongs to class C_1

Unit-2 Problems

(1) For the following data set predicts the output by linear regression model $w_1 x^t + w_0$. Calculate 'w1' and 'w0' using least Square estimates

Solution:

x^t	1	3	5	6
r^t	4	5	6	8

$\sum_t x^t = 15$, $\sum_t r^t = 23$, $N = 4$

$\sum_t x^t r^t = 97$, $\sum_t (x^t)^2 = 71$

$AW = y$

$$A = \begin{bmatrix} N & \sum_t x^t \\ \sum_t x^t & \sum_t (x^t)^2 \end{bmatrix} = \begin{bmatrix} 4 & 15 \\ 15 & 71 \end{bmatrix}$$

$$W = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} \quad y = \begin{bmatrix} \sum_t r^t \\ \sum_t x^t r^t \end{bmatrix} = \begin{bmatrix} 23 \\ 97 \end{bmatrix}$$

$$w_0 = \frac{\Delta_0}{\Delta} \qquad \Delta = \begin{vmatrix} 4 & 15 \\ 15 & 71 \end{vmatrix} = 59 \qquad \Delta_0 = \begin{vmatrix} 23 & 15 \\ 97 & 71 \end{vmatrix}$$

$$\Delta_0 = 178$$

$$w_0 = \frac{178}{59} = 3.0169$$

$$w_1 = \frac{\Delta_1}{\Delta} \qquad \Delta_1 = \begin{vmatrix} 4 & 23 \\ 15 & 97 \end{vmatrix} = 43$$

$$w_1 = \frac{43}{59} = 0.7288$$

$w_0 = 3.0169$

$w_1 = 0.7288$

3. Consider the following machine learning problem : (4M)

Kids Weight (Kgs)	20	22	24	26	28	30
Std I **ω1**	9	10	8	7	3	2
Std II **ω2**	2	3	6	9	9	9

Find the following for the above problem (Any 4 from below)

(a) Find $P(x|\omega_1)$ for x=24 Ans : 8/39
(b) Find $P(x|\omega_2)$ for x=24 Ans : 6/38
(c) Find $P(\omega_1|x)$ for x=24 Ans: 8/14
(d) Find $P(\omega_2|x)$ for x=24 Ans: 6/14
(e) Find p(x) for x=24 Ans : 14/77
(f) Find $P(x|\omega_1)/P(x|\omega_2)$ for x=24 Ans : (8*38)/(6*39)=152/117

4. Estimate the parameters (mean and variance) for the following class (4M)

Kids Weight (Kgs)	20	22	24	26	28	30
Std I **ω1**	9	10	8	7	3	2

Mean = (20*9+22*10+...30*2)/(9+10+...2)=23.53

Variance = 8.299

$$\sigma^2 = \frac{\sum(X-\mu)^2}{N}$$

THANK YOU

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